Benha University
Faculty of Engineering (Shoubra)
Electrical Engineering

ELE123 Technical Reports

ELE114 Technical Report Writing

(2026-2027)

Dr. Christina Gamal

Preparing Technical Reports Using MS Word

Lecture 3

Lecture Outlines

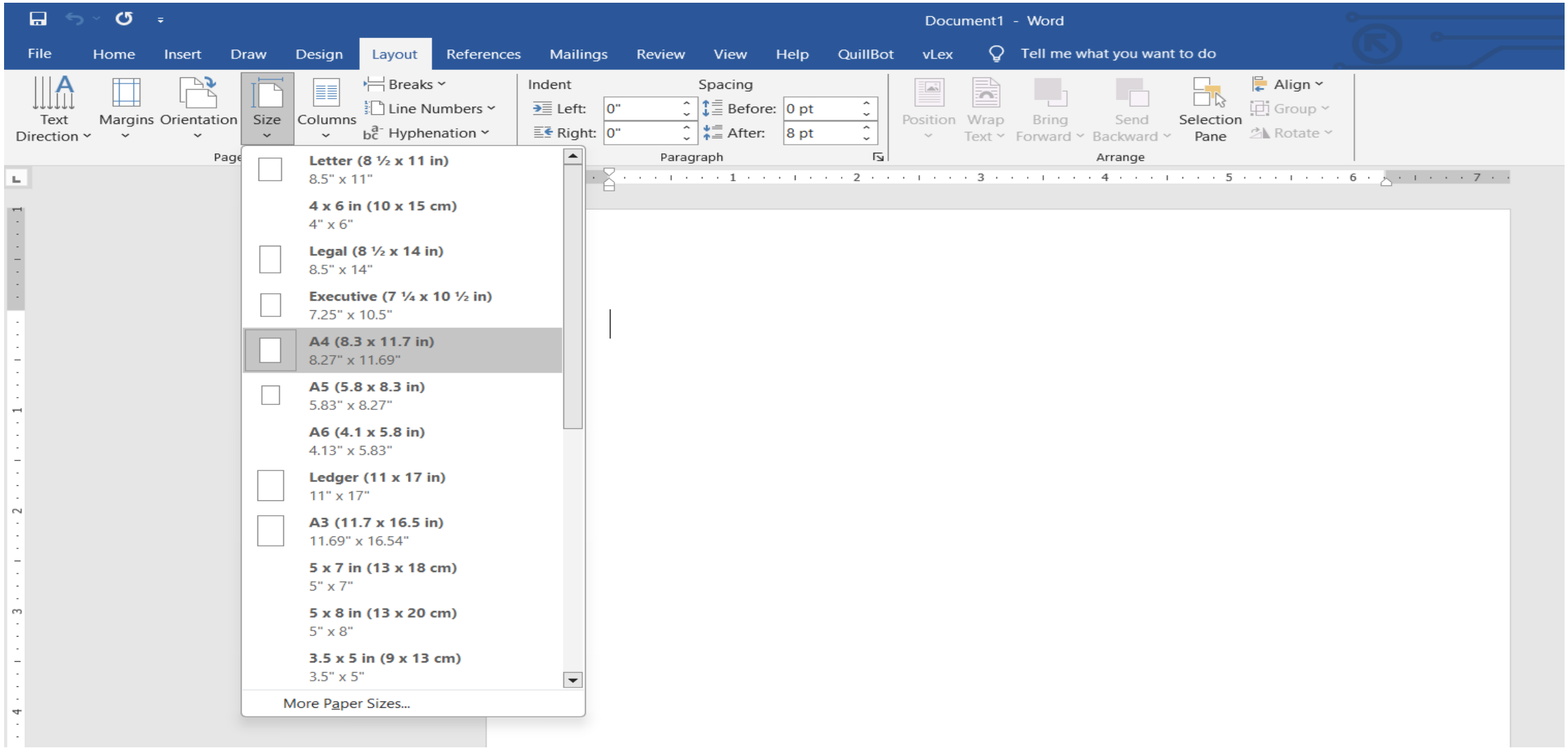
1- Paper Size and Margins
2- Cover Page
3- Page Numbering
4- Headers and Footers
5- Size, Font Style, and Format of Text
6- Heading and Table of Contents
7- Line Spacing
8- Indentation and Centering
19- Figures and Tables
10- Equations
11- Citations and References

1- Paper Size and Margins

- A4 paper is appropriate, and printing on one side only is often preferred for assessment purposes.
- Wide margins are recommended; for example, allowing 2.5 cm on all sides. Bound reports have a left-hand margin of at least 3.5 cm, and 1.5 cm on the other three sides.

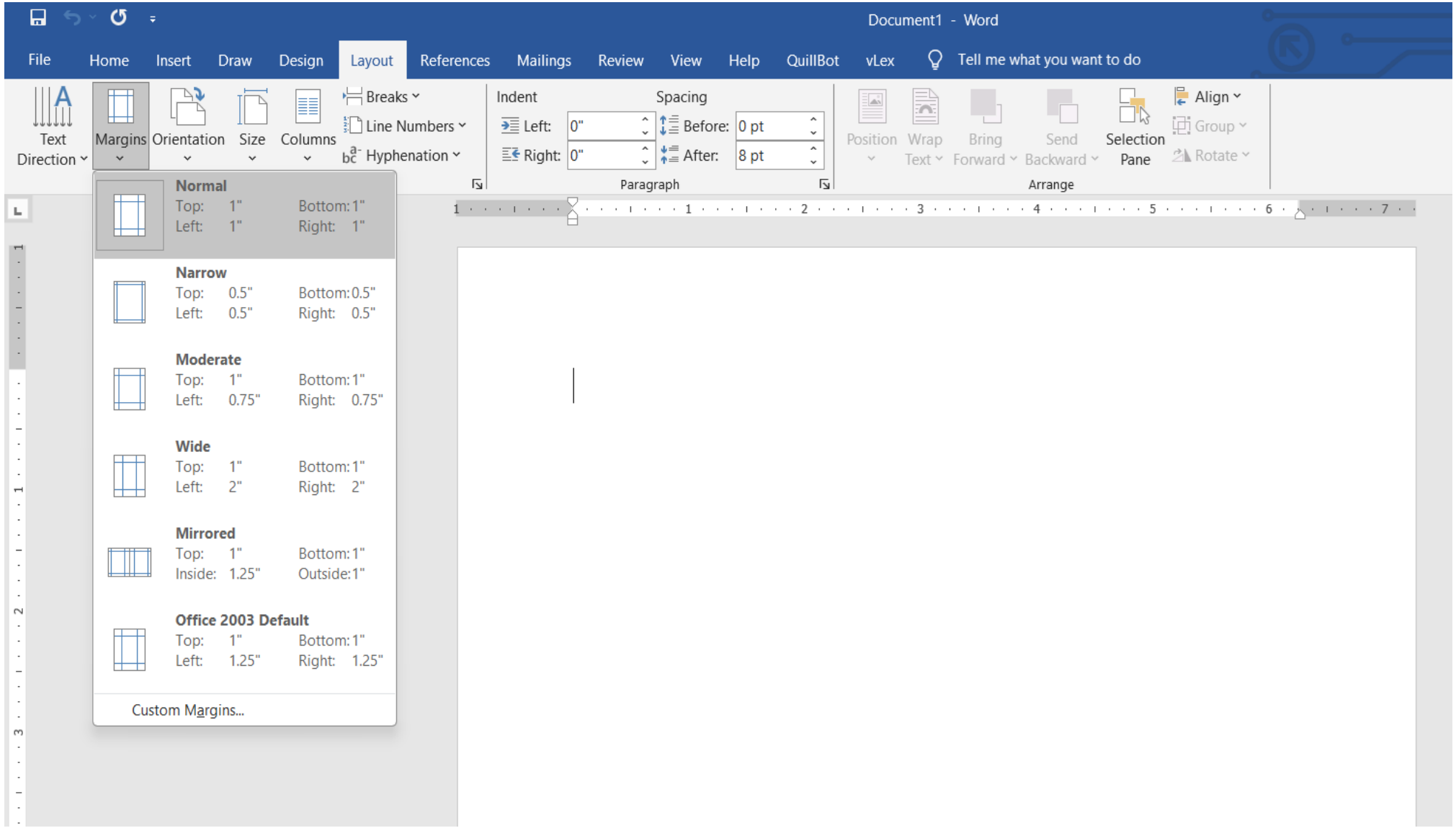
➢ How to adjust paper size in Word?

- For the paper size, click on **Layout**. Under tab **Size**, select **A4**.



➢ How to adjust Margins in Word?

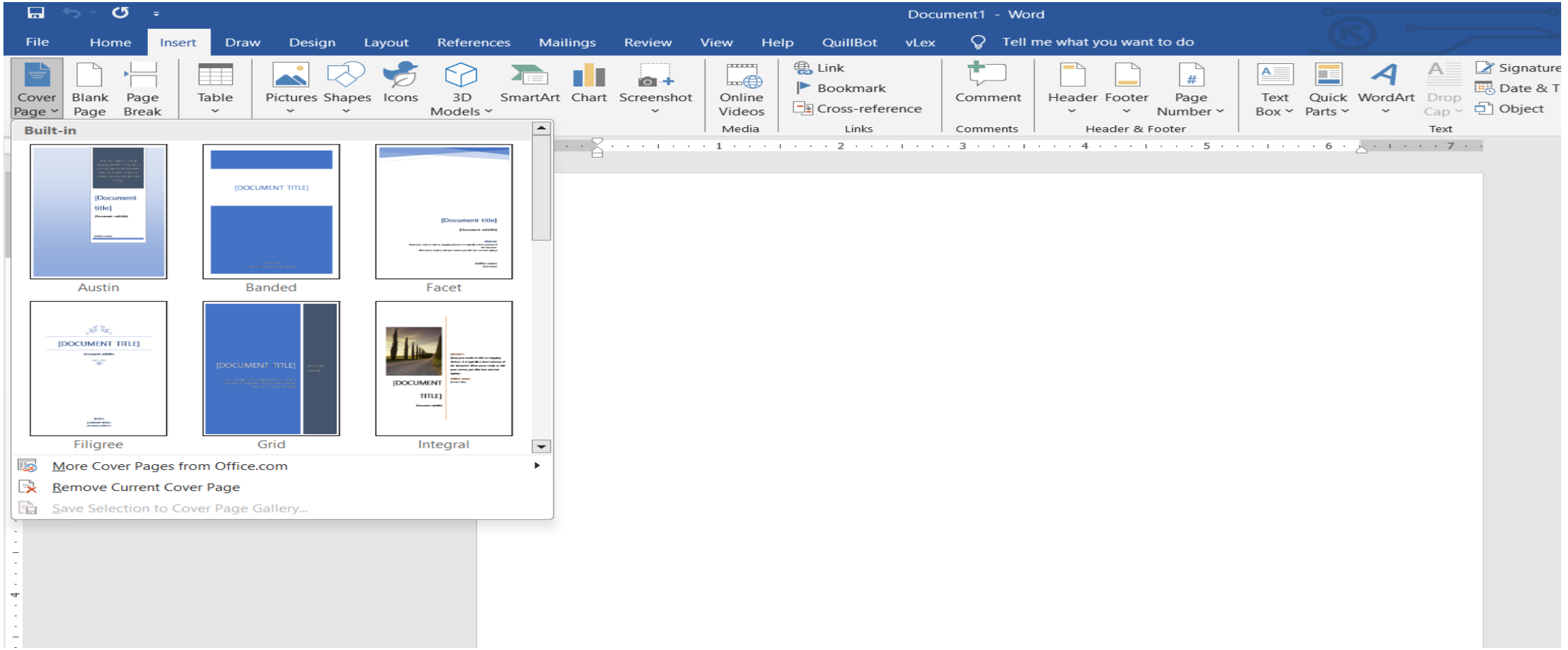
- For the page margins, click on **Layout**. Under tab **Margins**, select **Normal** margins.



2- Cover Page

➢ How to create cover page in Word?

- To create a cover page in Word, go to the **Insert tab**, click **Cover Page** in the Pages group, and select a design from the gallery.
- Then, click on the placeholder text fields and type your document's title, author, and date to customize the page.

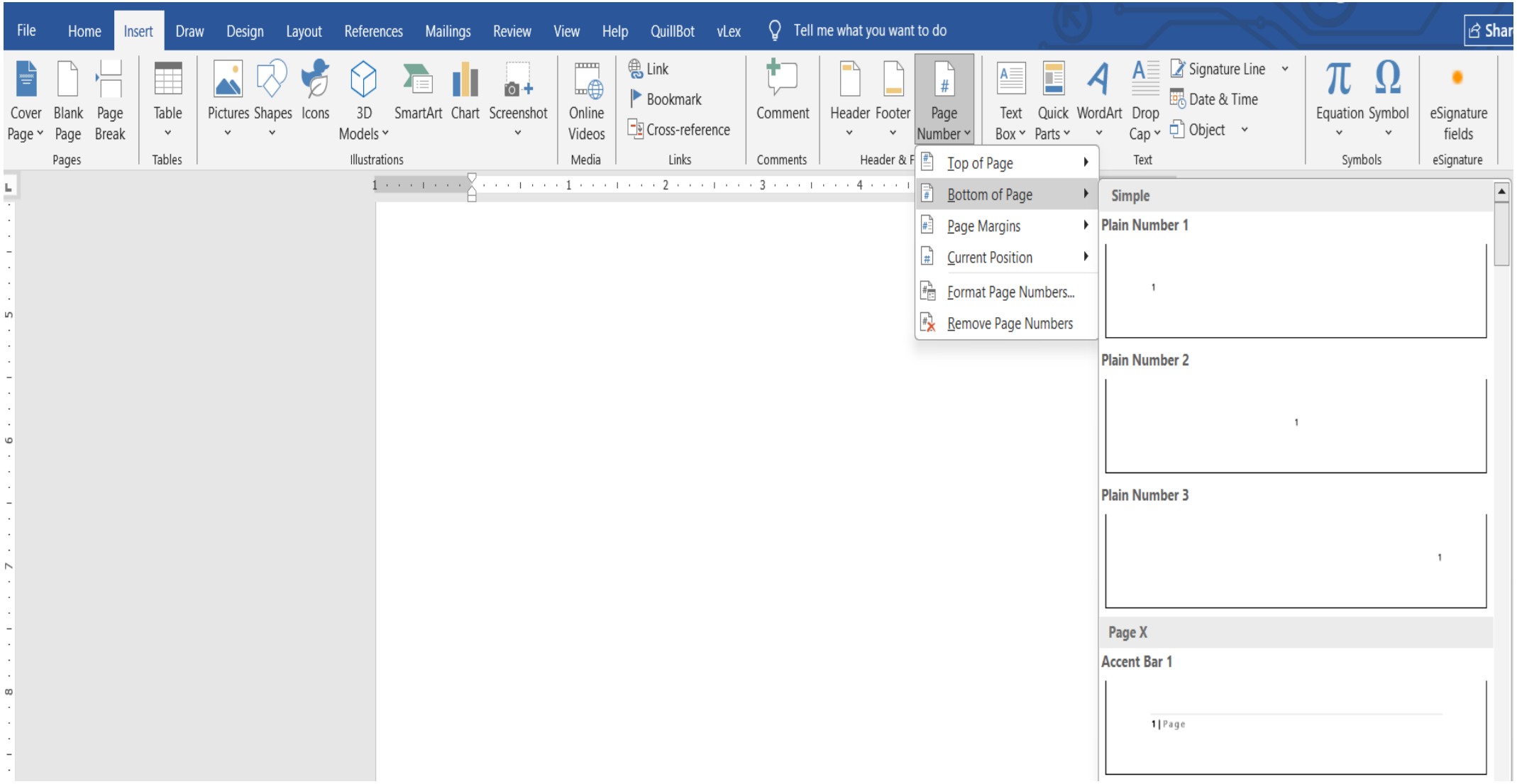


3- Page Numbering

- The title page should not be numbered.
- All other pages may be numbered either in the right-hand upper corner, or in the center at the bottom of the page.
- Roman numerals — i, ii, iii, iv etc. — are usually used to number the preliminary pages.
- Arabic numerals — 1, 2, 3, 4 etc. — are used to number the main text.
- Pages in the Appendices can be numbered internally, according to the letter of the individual appendix and the number of pages within each appendix — A1, A2, B1, C1, C2, C3 etc. (where Appendix A has two pages; Appendix B has one page, and so on).

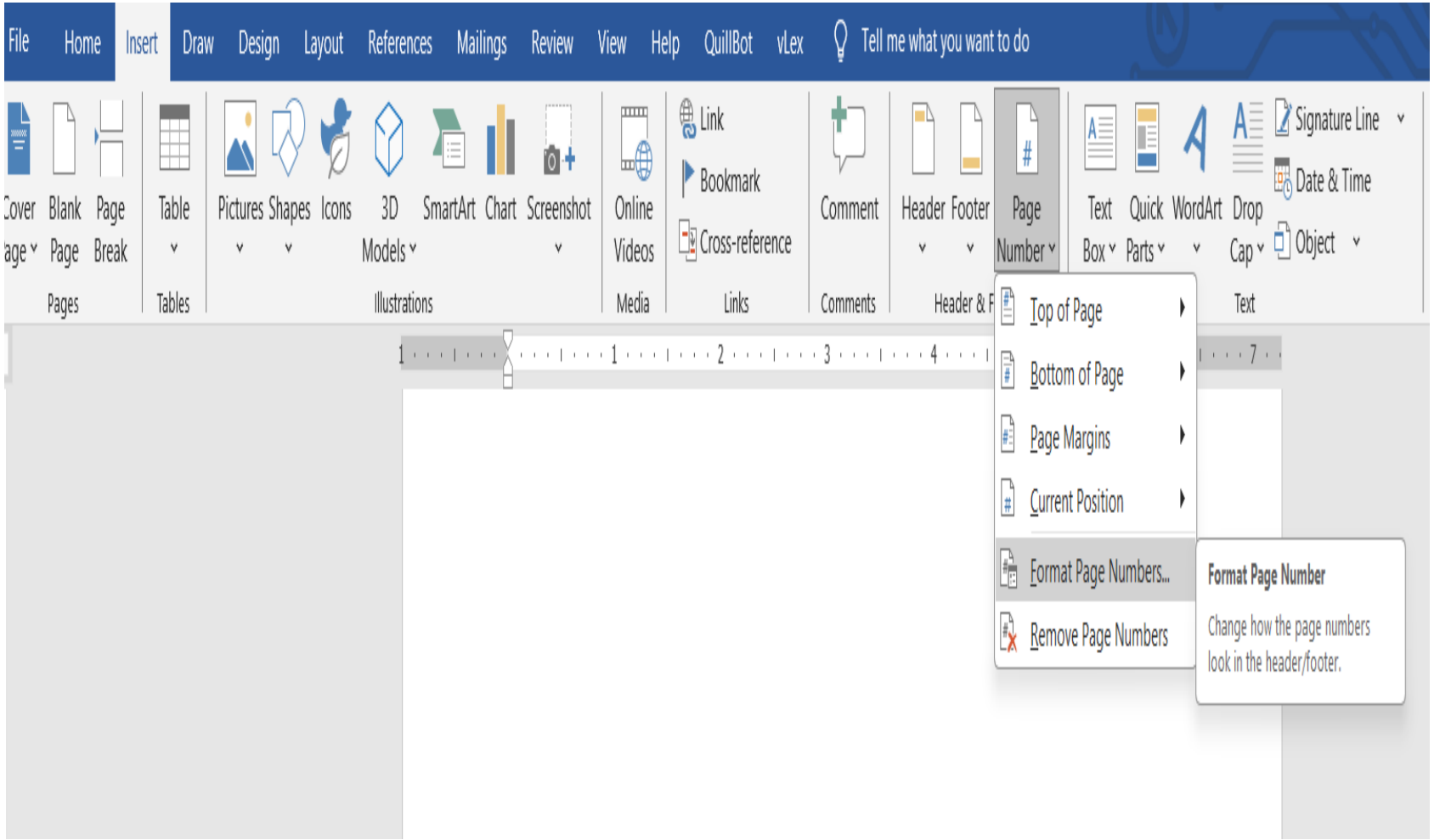
➢ How to format page numbers in Word?

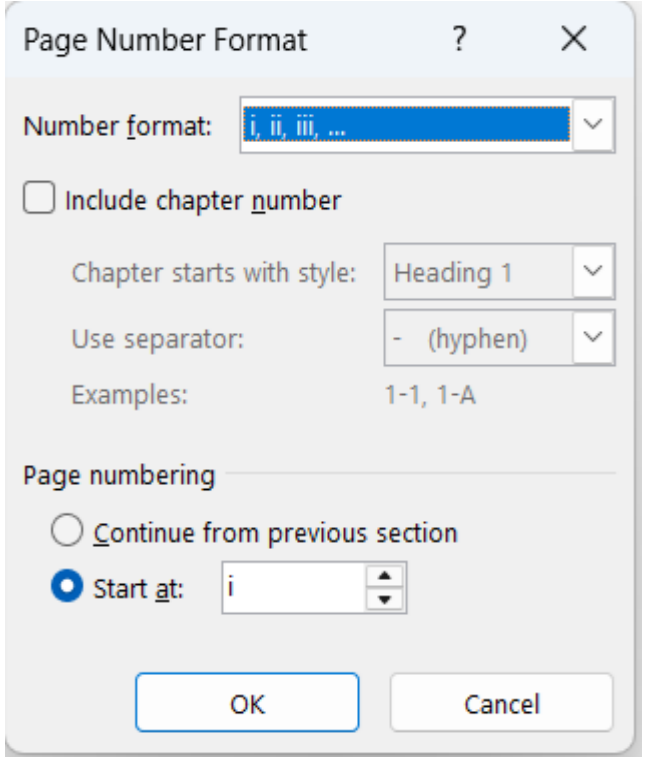
- For the page numbers, click on **Insert**. Under tab **Page Number**, select **(Top of Page** or **Bottom of Page)** to control the position of the page numbers.



➢ How to format page numbers in Word? …. Cont.

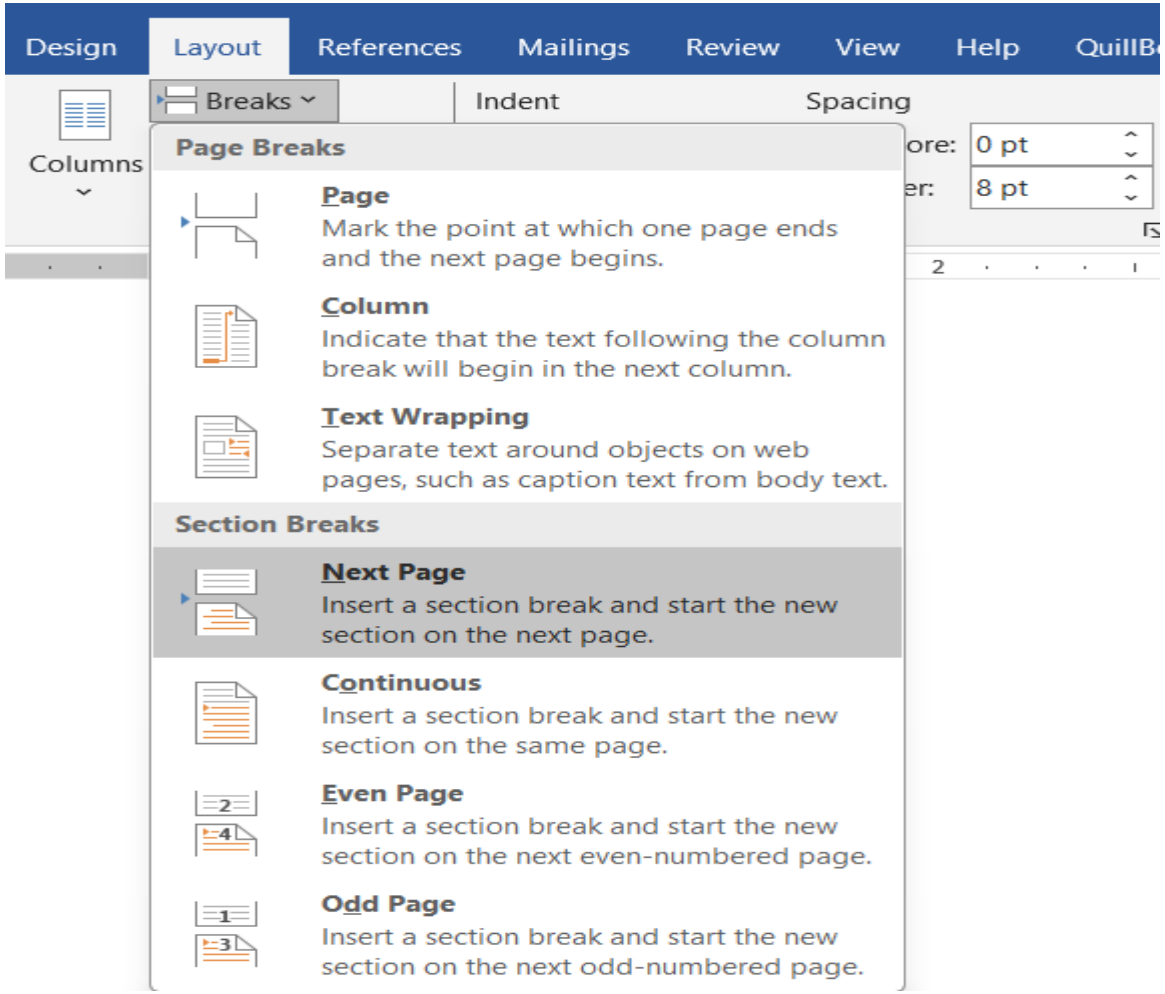
- Then from tab **Page Number**, select **Format Page Numbers** to select the number format (Roman numerals, Arabic numerals,….)

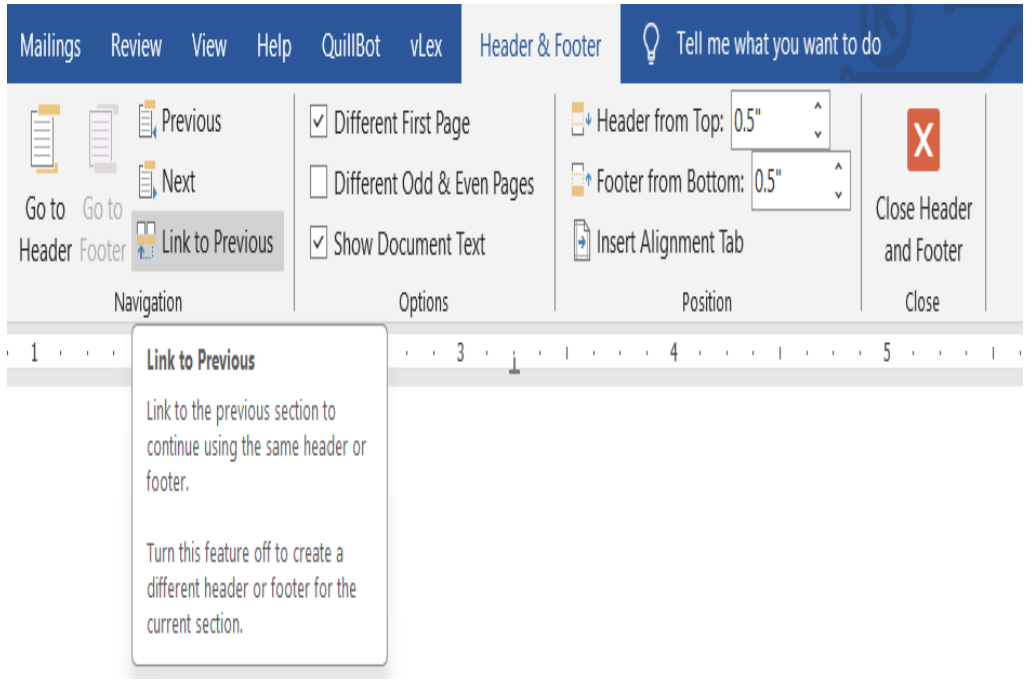




➢ How to format page numbers in Word? …. Cont.

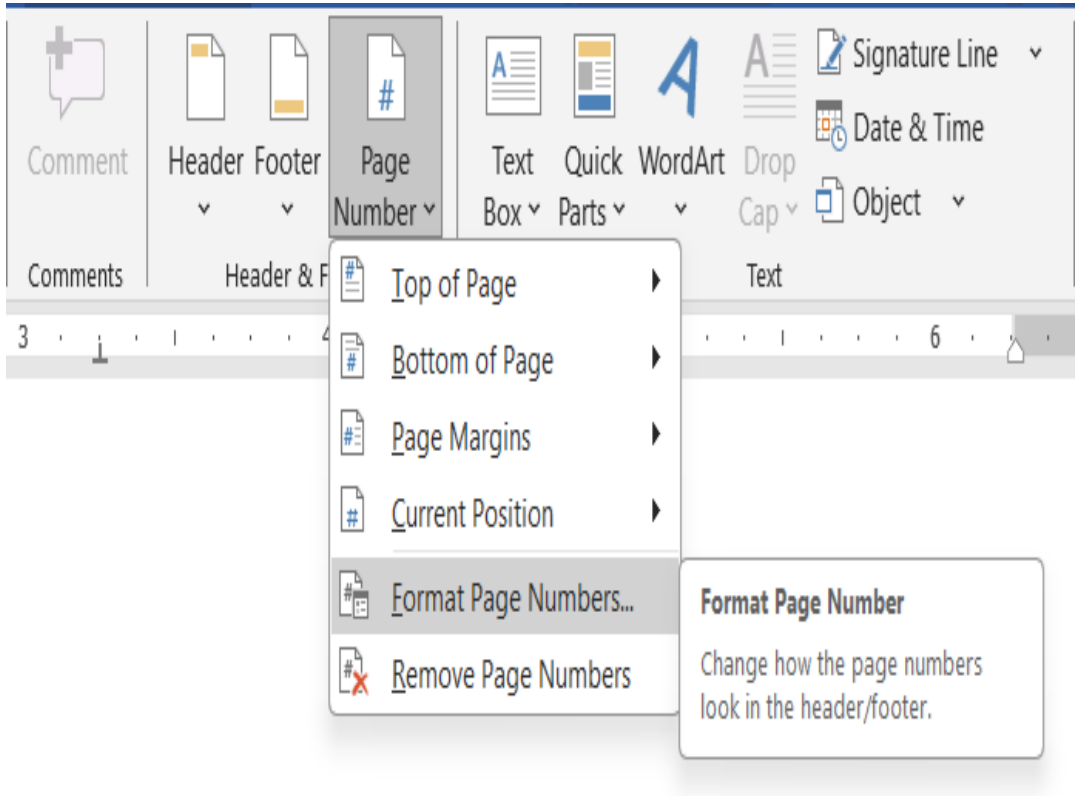
- To switch to Roman numerals and set the starting number (i) for the front matter, and then set it to "1" for the main body, you'll insert **section breaks** in your document, turn off the **"Link to Previous"** function in headers/footers to create separate sections.

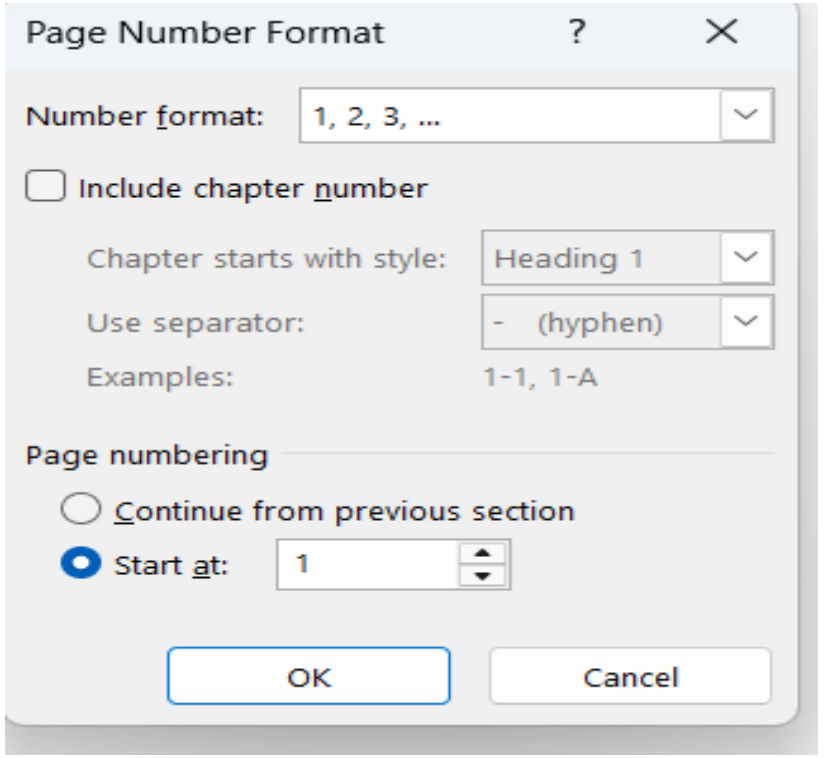




➢ How to format page numbers in Word? …. Cont.

- Then use the **“Format Page Numbers”** option.



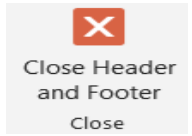


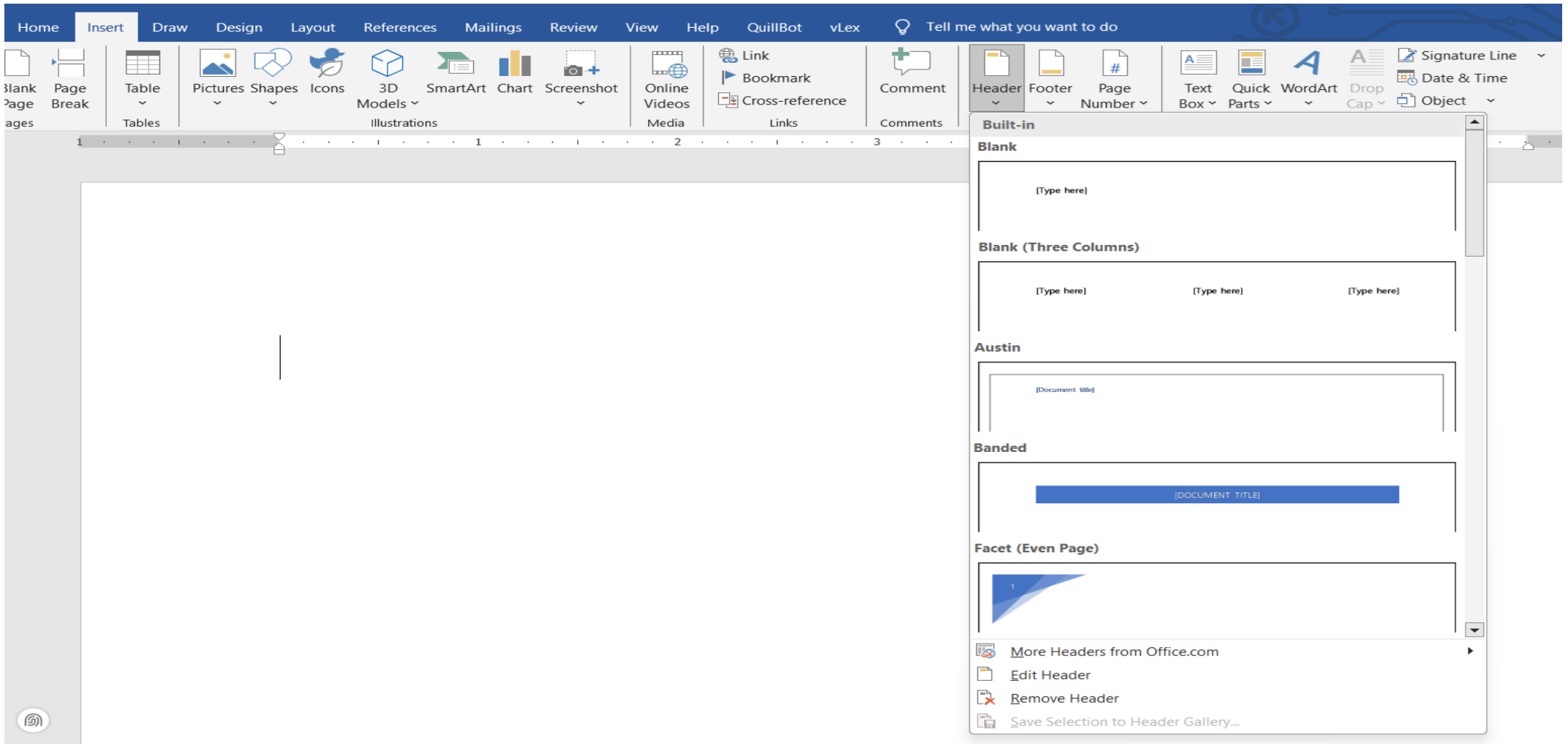
4- Headers and Footers

- Headers or footers position page numbers automatically.
- Two different headers or footers must be created so that the preliminary pages are numbered separately from the text of the report.
- Student name and ID number can appear in a footer in 9-point size.

➢ How to insert Header and Footer in Word?

- To insert a header or footer in Word, click the **Insert tab**, then select either **Header or Footer** and choose a preset option, or select Edit Header/Footer to create your own.
- You can type your custom text, add elements like page numbers or the date, and then click Close Header and Footer or press the Esc key to exit the header or footer editing view.





5- Size, Font style, and Format of text

❖ Size of text

- 12-point type is the normal size for the text of a report. 10-point may be used for figures and tables.
- Headings without text should never appear on the bottom line of a page.
- Technical reports use headings to divide information into sections. The headings help reader locate relevant information quickly.

First-level	16 bold	**6 Environmental Impact**
Second-level	14 bold	**6.1 Chemical Waste**
Third-level	12 bold	**6.1.1 Cost Analysis**
Fourth-level	12 bold italics	***6.1.1.1 Equipment Costs***

- Capital letters are used for the first letter of the first word in each heading only; except for any acronyms (e.g. IEEE), trade names, or personal names and places, which may require more than one capital.

5- Size, Font style, and Format of text … Cont.

❖ Font style of text

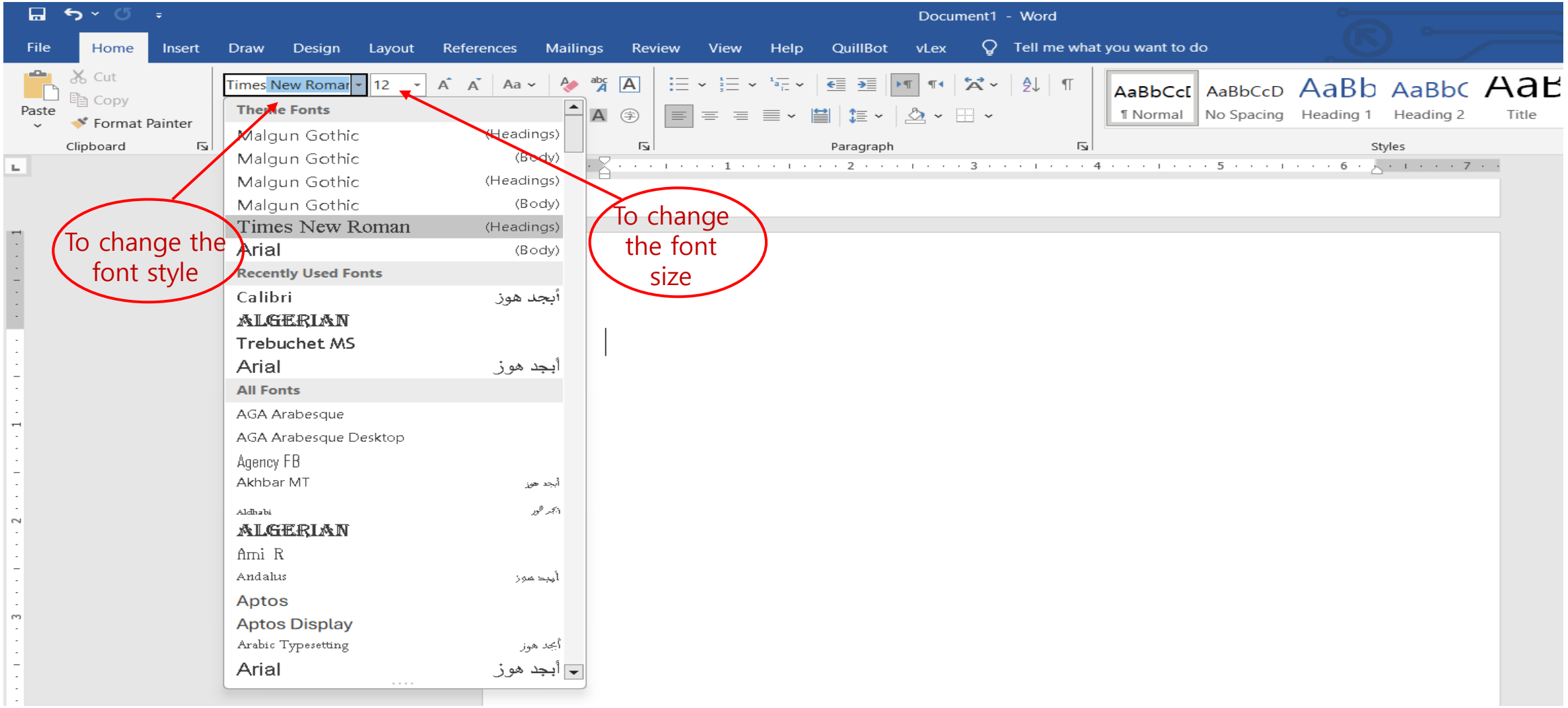
- A Times New Roman is usually chosen for the text of a report.
- A different font (such as Arial) may be used for headings and tables.
- The same font should be used throughout the whole report, unless a second font is chosen for headings and tables.

❖ Format of text: Justification

- Text must be justified (right and left margins).

➢ How to select Size and Font in word?

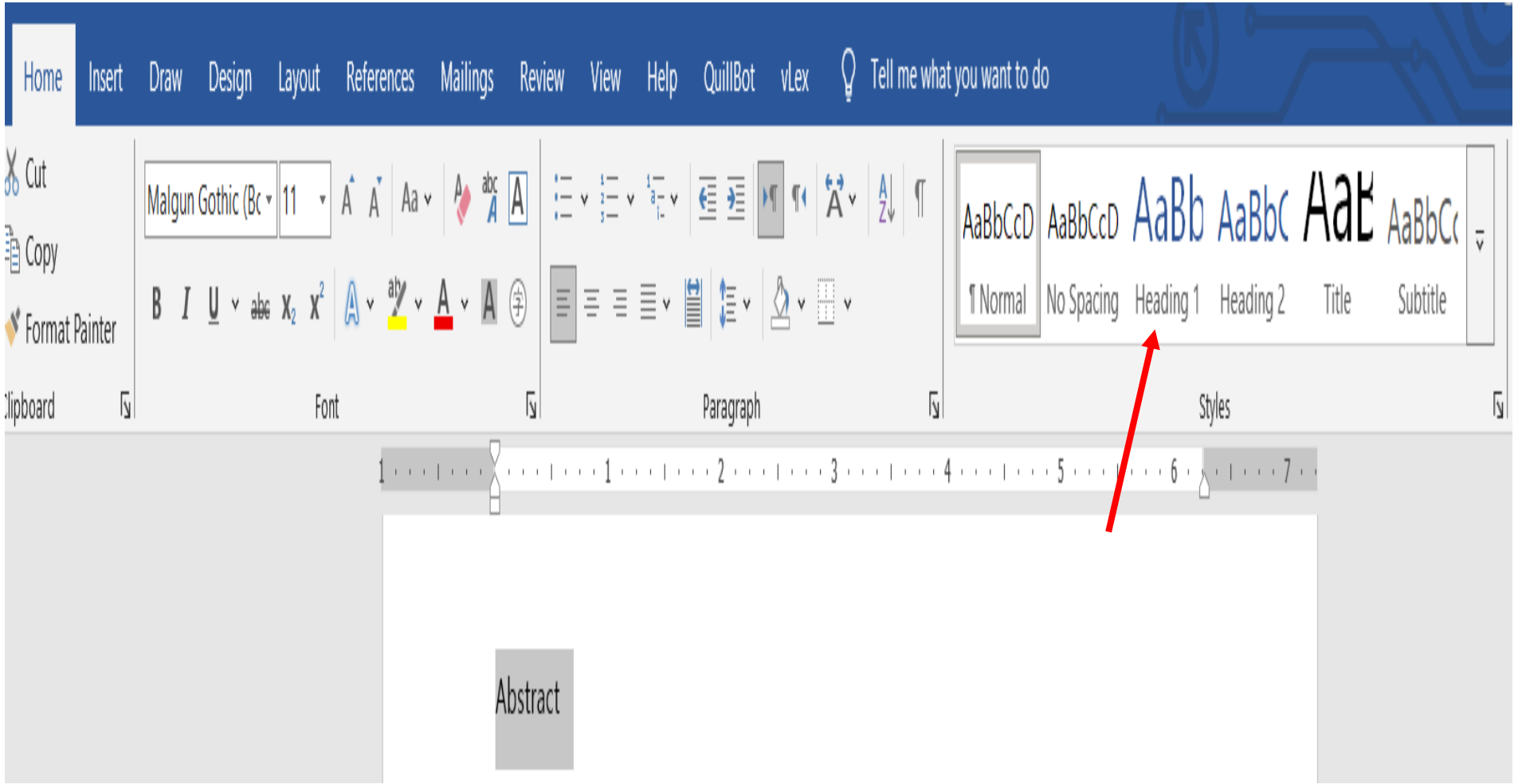
- First select the text you want to change. Then, go to the **Home tab**, locate the Font group, and use the dropdown menus to choose a new font style and font size.



6- Heading and Table of Contents

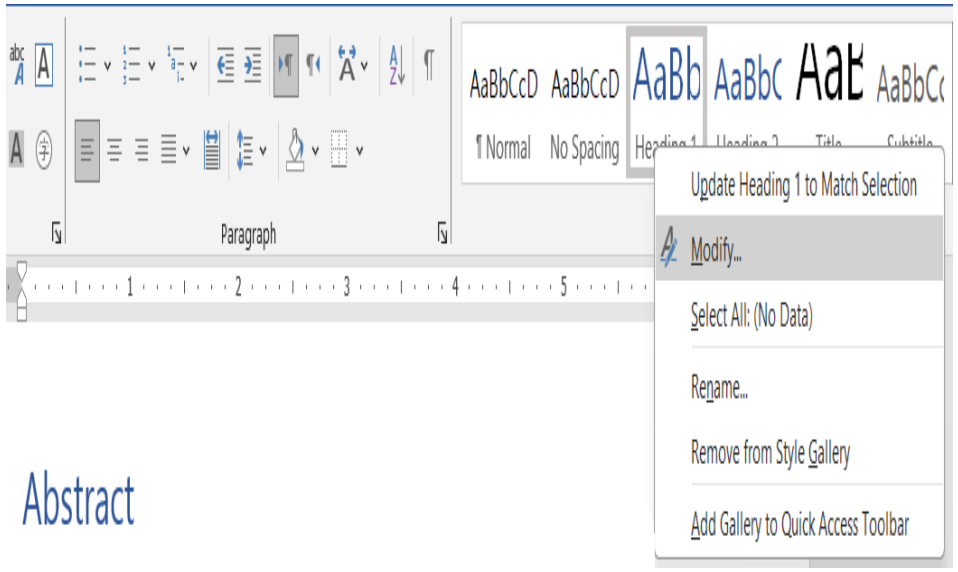
➢ How to modify Heading styles in Word?

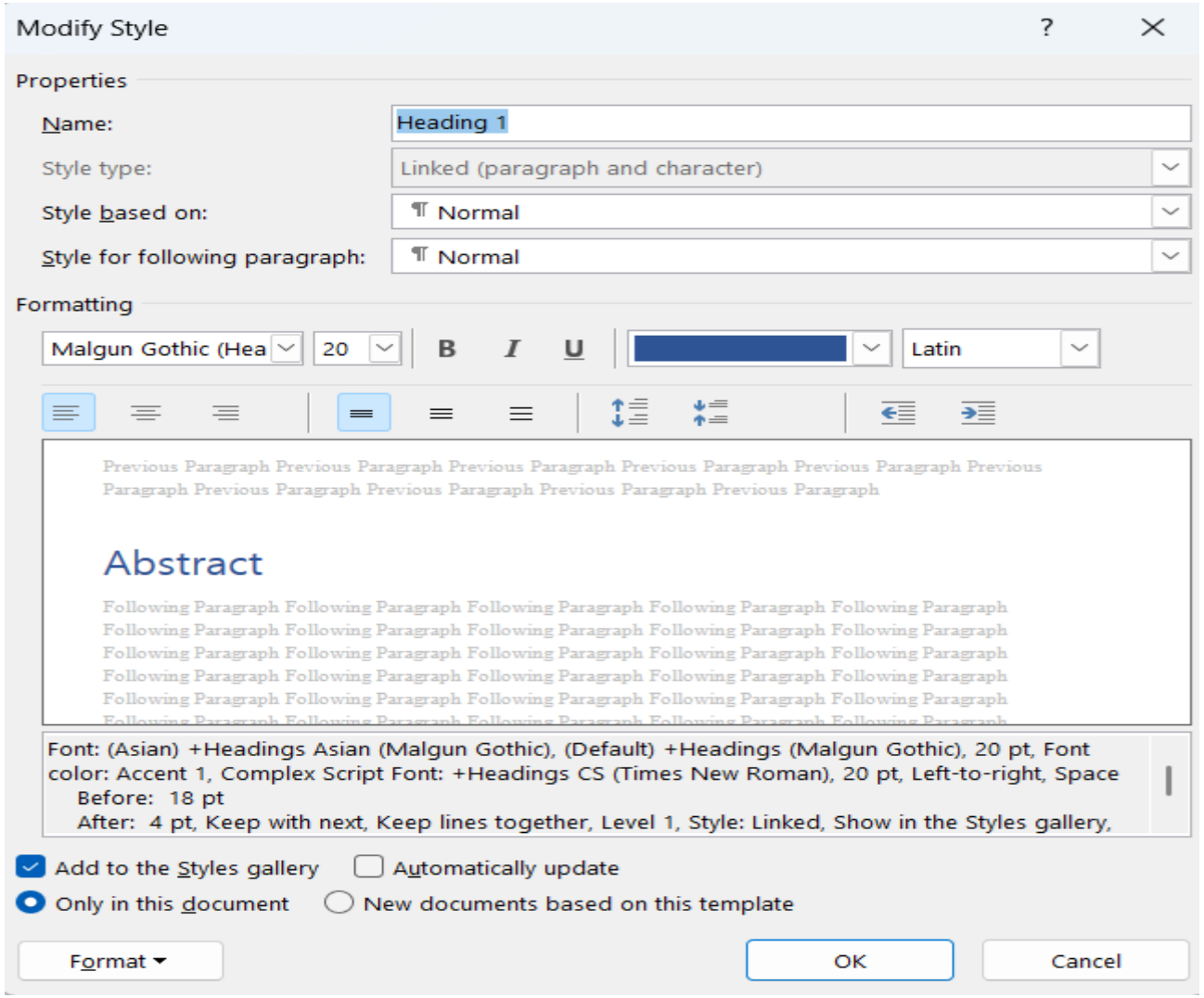
- To modify heading styles in Word, go to the Home tab, find the Styles group, right-click the desired heading style (e.g., Heading 1), and select Modify from the menu.



➢ How to modify Heading styles in Word? … Cont.

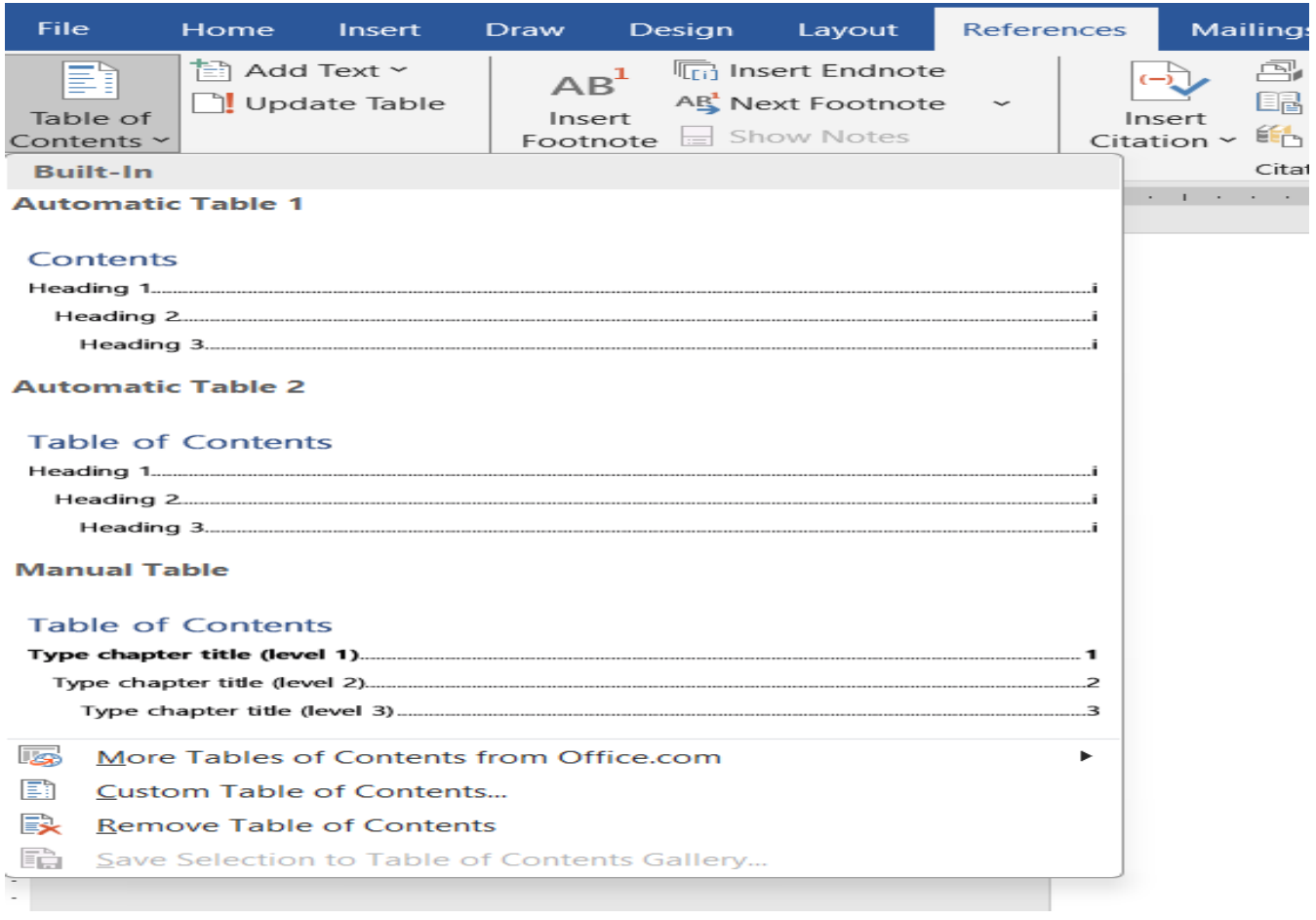
- In the dialog box, use the formatting tools to change the font, color, size, or paragraph settings as desired, then click OK to update the style throughout the document.




Modify Style
Properties
Name:
Heading 1
Style type:
Linked (paragraph and character)
Style based on:
Normal
Style for following paragraph:
Normal
Formatting
Malgun Gothic (Hea
20
B
I
U
Latin
Previous Paragraph Previous Paragraph Previous Paragraph Previous Paragraph Previous Paragraph Previous Paragraph Previous Paragraph Previous Paragraph Previous Paragraph Previous Paragraph Previous Paragraph
Abstract
Following Paragraph Following Paragraph Following Paragraph

➢ How to create Table of Contents in Word?

- To create a Table of Contents (TOC) in Word, you must first format your document's headings using Word's built-in Heading styles (Heading 1, Heading 2, etc.) on the Home tab.
- Then, go to the **References tab**, click **Table of Contents**, and select an automatic TOC style or use Custom Table of Contents to specify details.
- Word automatically populates the TOC with the text and page numbers from your styled headings.



7- Line Spacing

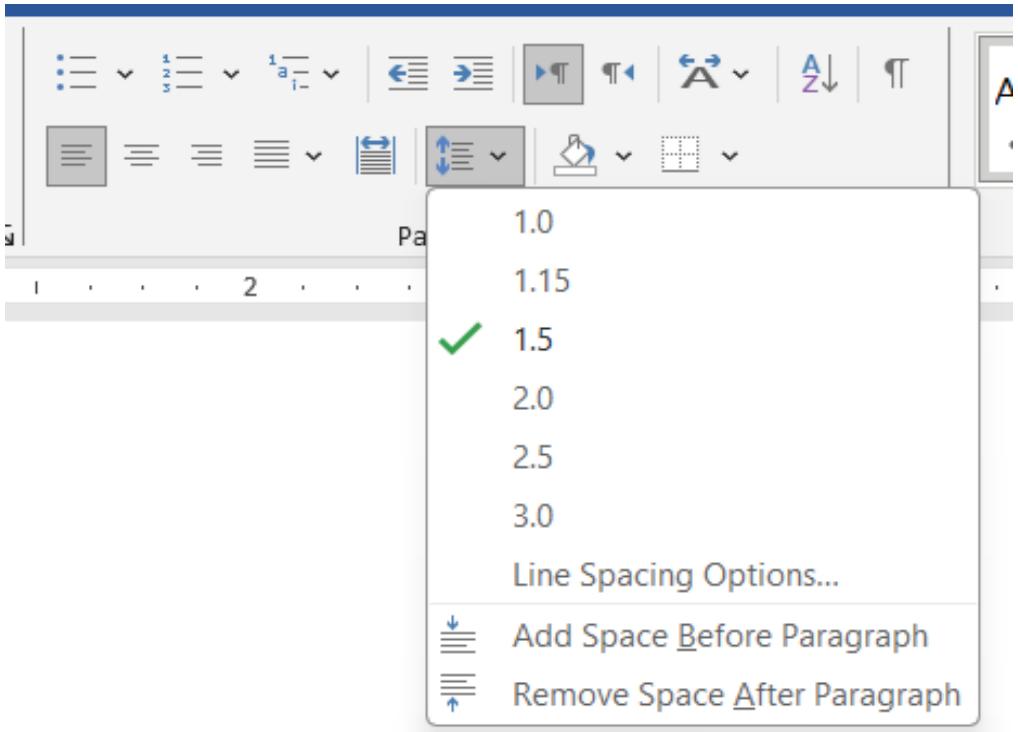
1.5 line spacing is generally used for work submitted for assessment.

Extra line spacing should be used in the following cases :–

- to separate paragraphs
- to separate figures, tables, equations, and long quotations from the text
- to separate a main heading from the section of text which comes before

➢ How to select line spacing in Word?

- First, **highlight the text** you want to format. Then, go to the **Home tab**, click the **Line and Paragraph Spacing** icon in the Paragraph group, and choose a preset spacing option (like single, 1.5, or double). For more control, select **Line Spacing Options** to open the Paragraph dialog box, where you can set custom spacing and adjust space before or after paragraphs.

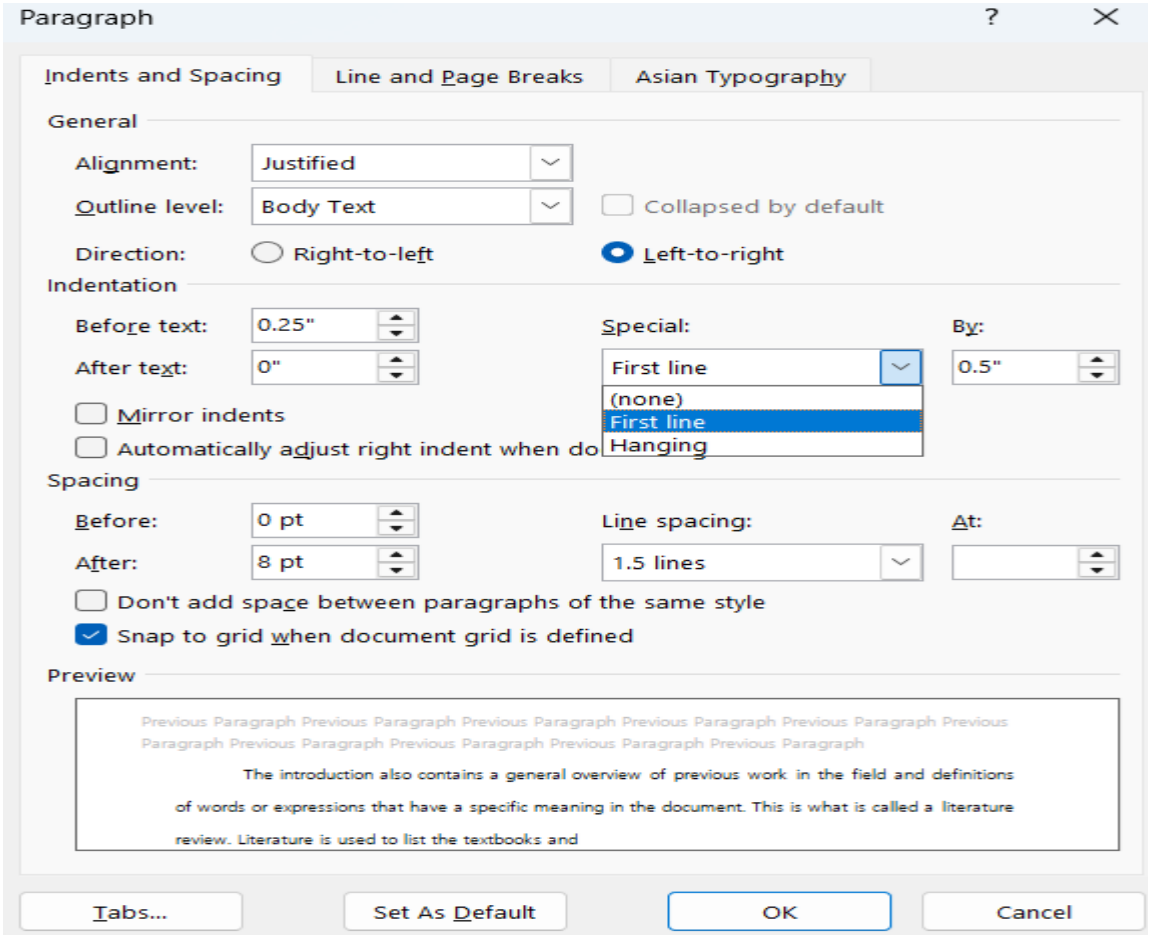


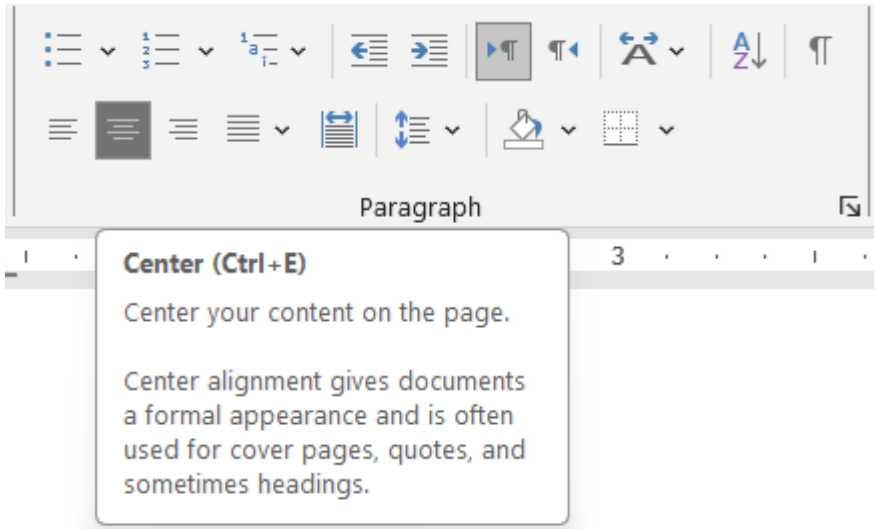
8- Indentation and Centering

- Indenting text at the beginning of a paragraph is necessary. But if an extra line space is inserted between paragraphs, no need for the indentation.
- Indentation is appropriate for lists with bullet points (•)
- Figures and tables may be centered in the text

➢ How to apply indentation and centering in Word?

- To apply indentation in Word, select your text, go to the **Home tab**, and click the **Paragraph dialog box launcher** in the Paragraph group to open the Paragraph dialog box. Under the "Indents and Spacing" tab, use the "Special" options (First Line, Hanging) to set your desired indent.
- To center figure or table, select it and click the **Center alignment** icon in the Paragraph group on the **Home tab.**





9- Figures and Tables

- Figures include diagrams, graphs, sketches, photographs, and maps
- Tables summarize data in rows and columns; a dash (—) indicates no data
- All figures and tables may be centered
- All figures and tables should be labelled in bold with the appropriate consecutive number, (figures are generally labelled at the bottom, and tables at the top)
- All figures and tables must be referred to in the text (e.g. ‘see Figure 14’); avoid using words such as ‘Figure above’ or ‘Table below’ Figures and tables can be separated from the text with extra line spacing.

Example:

The single-diode model, shown in Fig. 1, has become the most widely used model as it offers a good compromise between simplicity and accuracy [1,2].

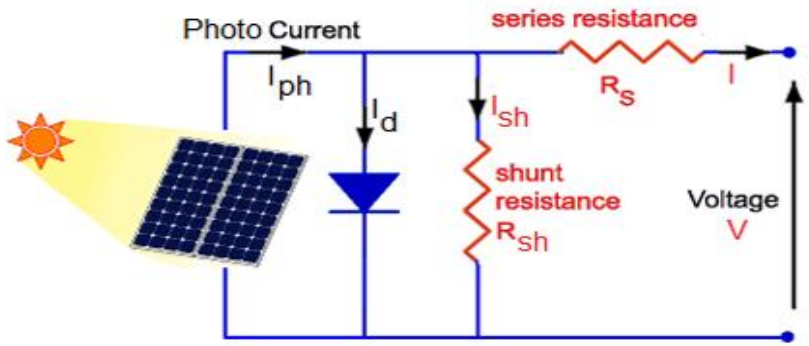


Fig. 1. Single-diode equivalent circuit for PV module

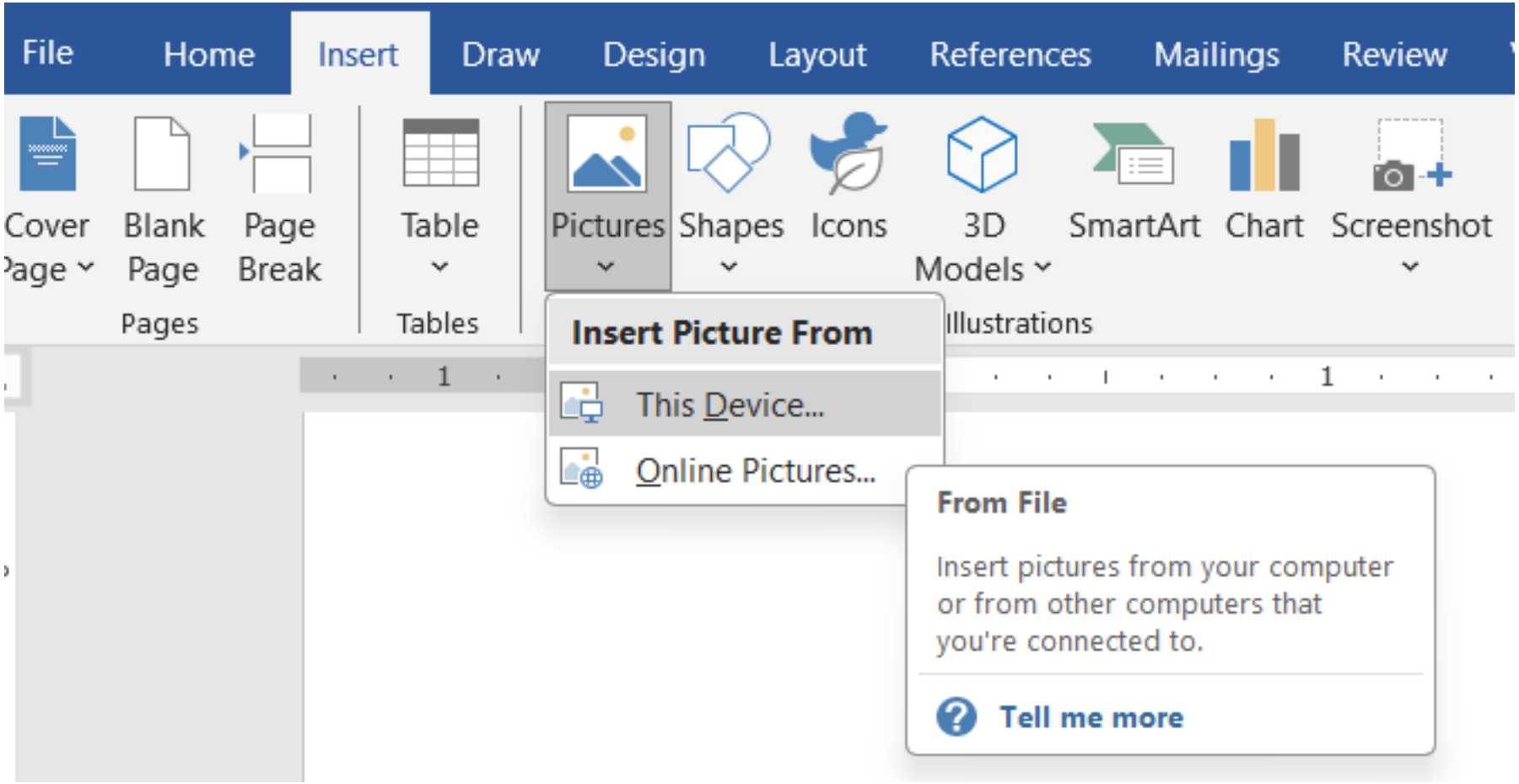
Two examples of such module are BP SX-150 and MSX-60 modules whose basic information are presented in Table 1. More details about these two PV modules are provided by manufacturer websites.

Table 1: Electrical characteristics of BP SX-150 and MSX-60 PV modules at STC

Parameters	Datasheet Value	
	BP SX-150	MSX-60
Maximum Power (P_{max})	150 W	60 W
Voltage at MPP (V_{mp})	34.5 V	16.8 V
Current at MPP (I_{mp})	4.35 A	3.56 A
Short-circuit current (I_{sc})	4.75 A	3.87 A
Open-circuit voltage (V_{oc})	43.5 V	21.0 V
Temperature coefficient of I_{sc}	(0.065±0.015) %/°C	(0.065±0.015) %/°C
Temperature coefficient of V_{oc}	– (160±20) mV/°C	– (80±10) mV/°C
Series-connected cells (N_s)	72	36

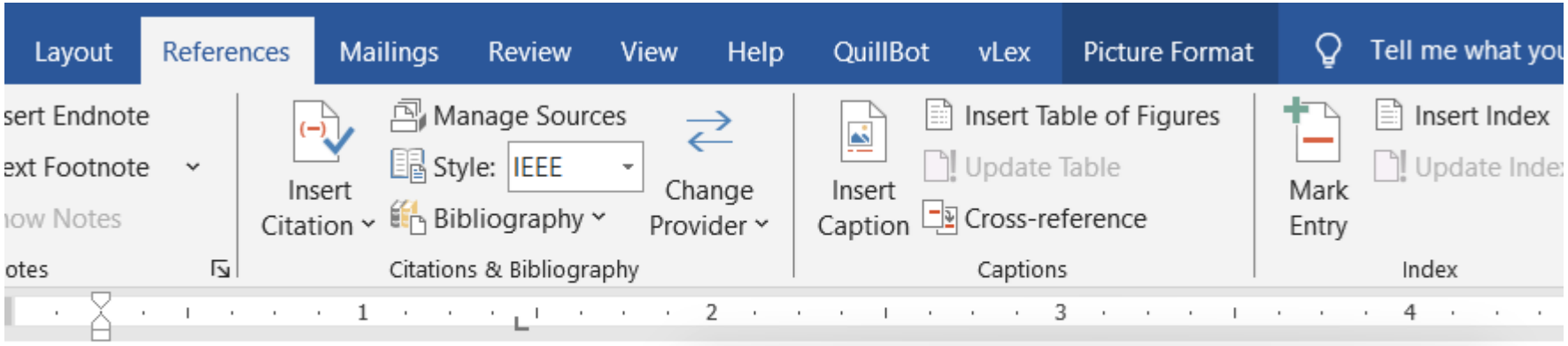
➢ How to insert Figure in Word?

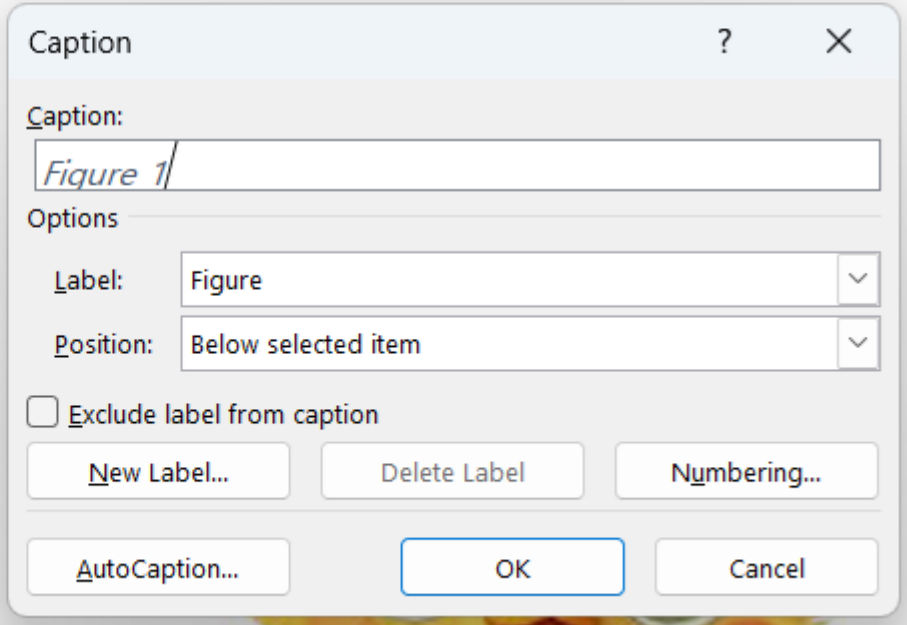
To insert a figure in Word, navigate to your desired insertion point, go to the **Insert tab**, click **Pictures**, choose your image file, and then **click Insert**.



➢ How to insert caption to Figure in Word?

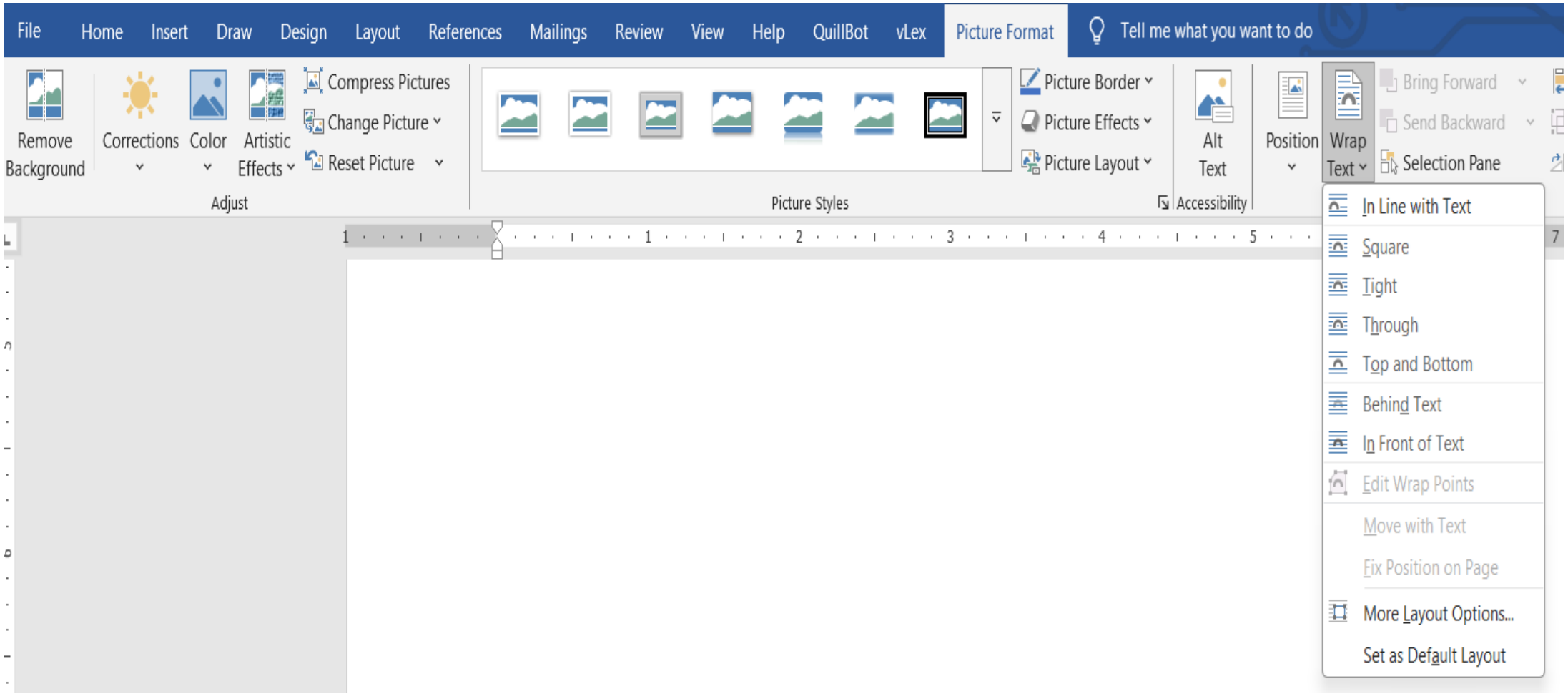
First **select the figure**, then go to the **References tab** and click **Insert Caption** to add a descriptive title with an automatic number.





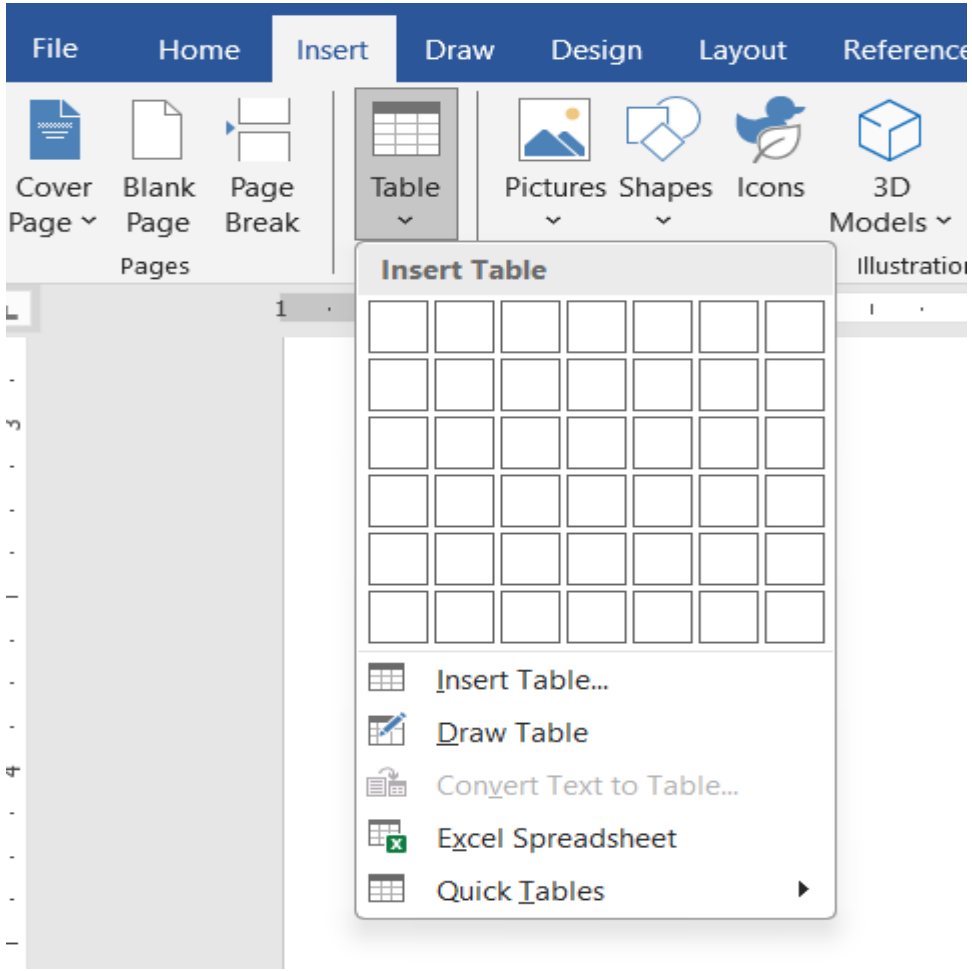
➢ How to adjust the Figure's position with text in Word?

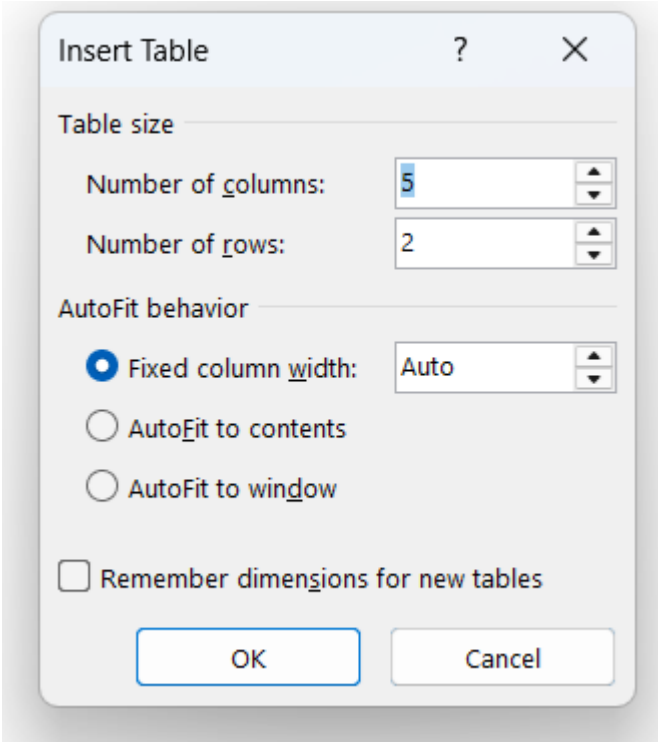
- To reposition a figure, place your cursor in the figure and go to the **Picture Format tab** and click **Wrap Text** to change how text wraps around it.



➢ How to insert Table in Word?

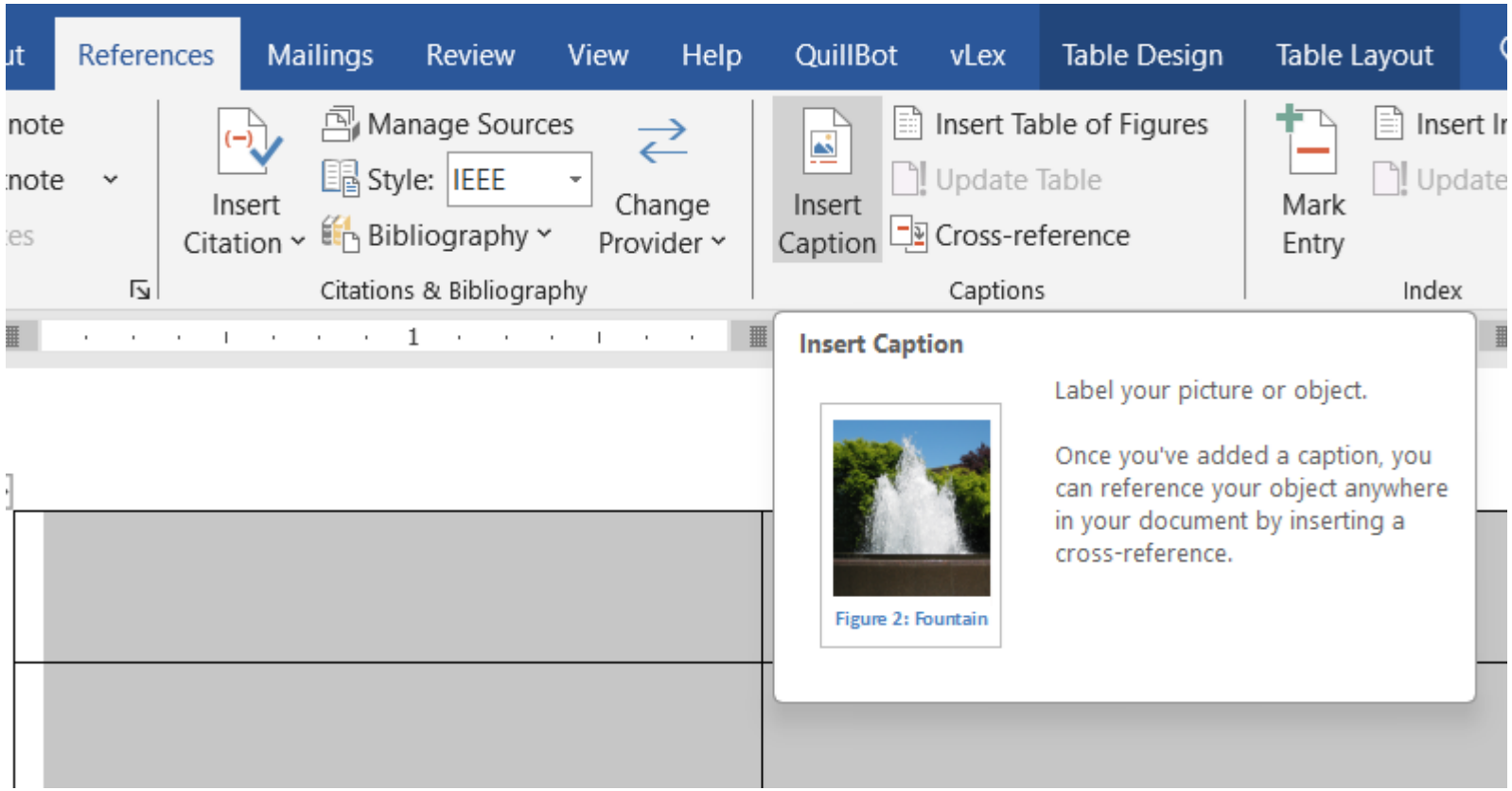
To insert a table, click on the **Insert tab**, then select Table and choose your desired number of rows and columns using the grid, or select **Insert Table** to specify dimensions in a dialog box.

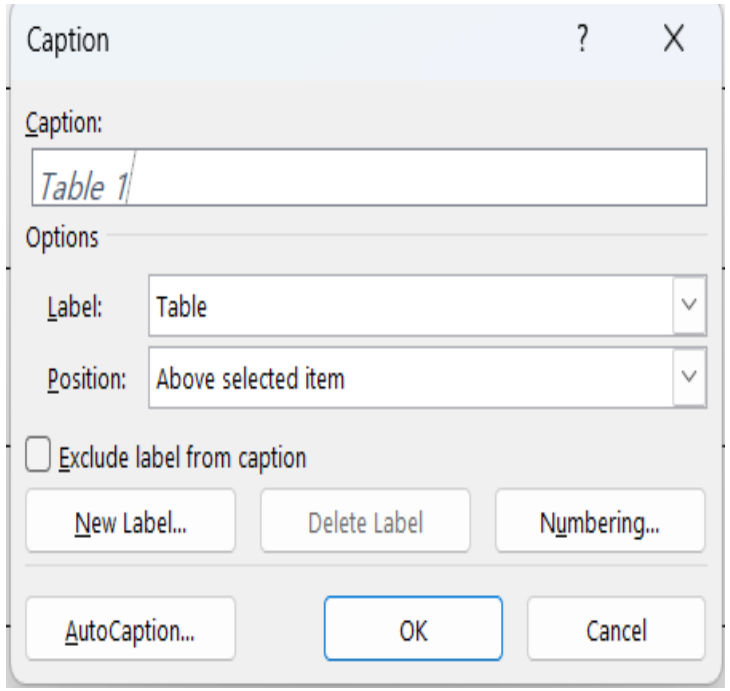




➢ How to label Table in Word?

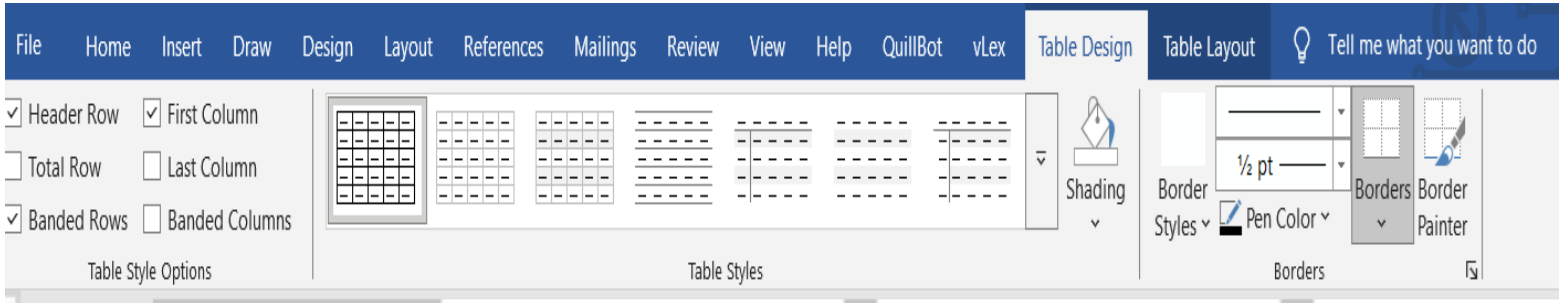
To label a table in Word, **select the table**, then go to the **References tab** and click **Insert Caption**. In the dialog box, select Table from the Label list, add your desired title after the label and number, and choose the position (usually above the table). Click OK to insert the labeled table.

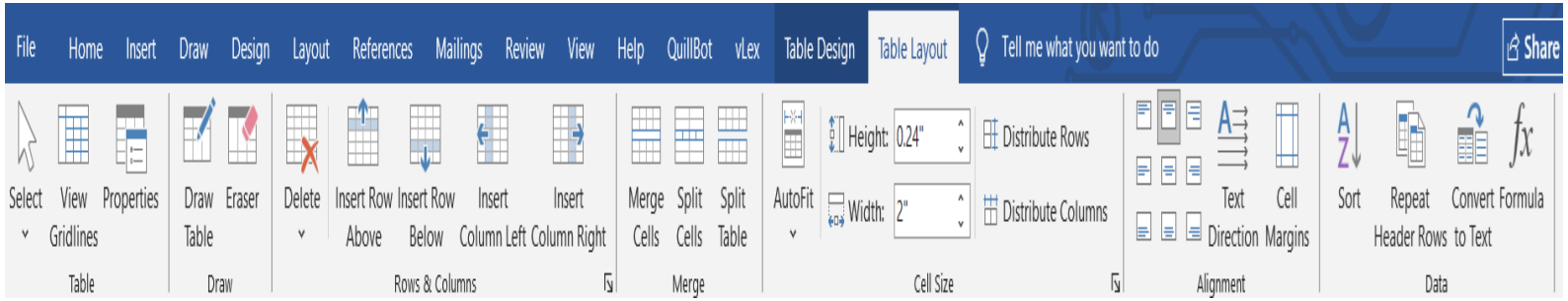


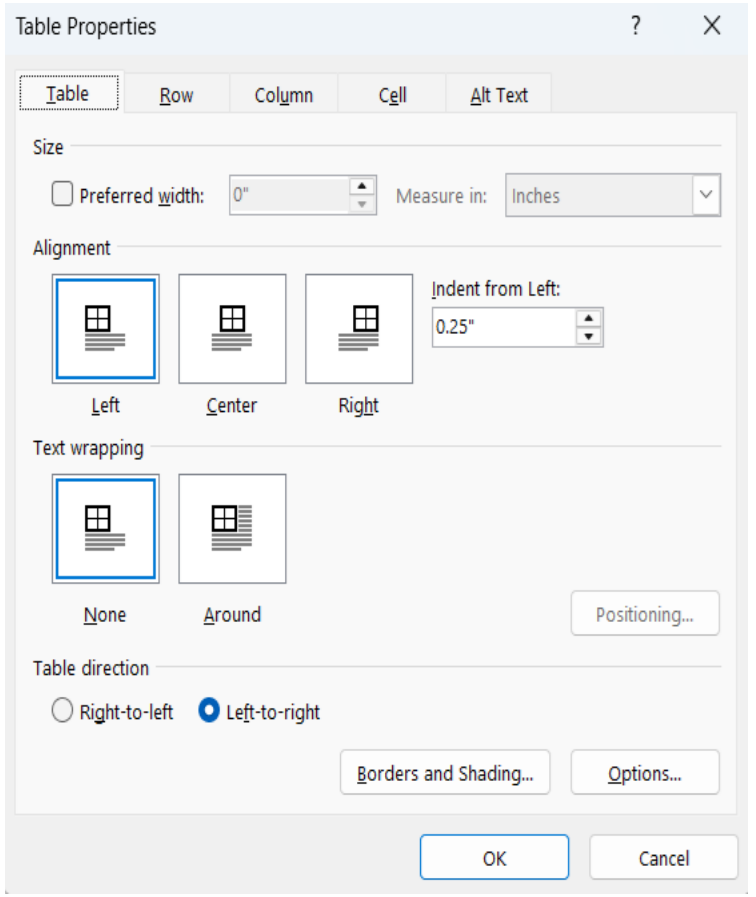


➢ How to format Table in Word?

- To format a table in Word, place your cursor in the table and go to the **Table Design** tab to apply pre-made table styles, or customize the style with options for headers, banded rows, and more.
- You can also use the **Table Design** tab to change borders and shading, or use the **Table Layout** tab to adjust cell size, Merge and text alignment. For more advanced formatting, use the **Table Properties dialog** to control table positioning, text wrapping, and alignment with surrounding text.

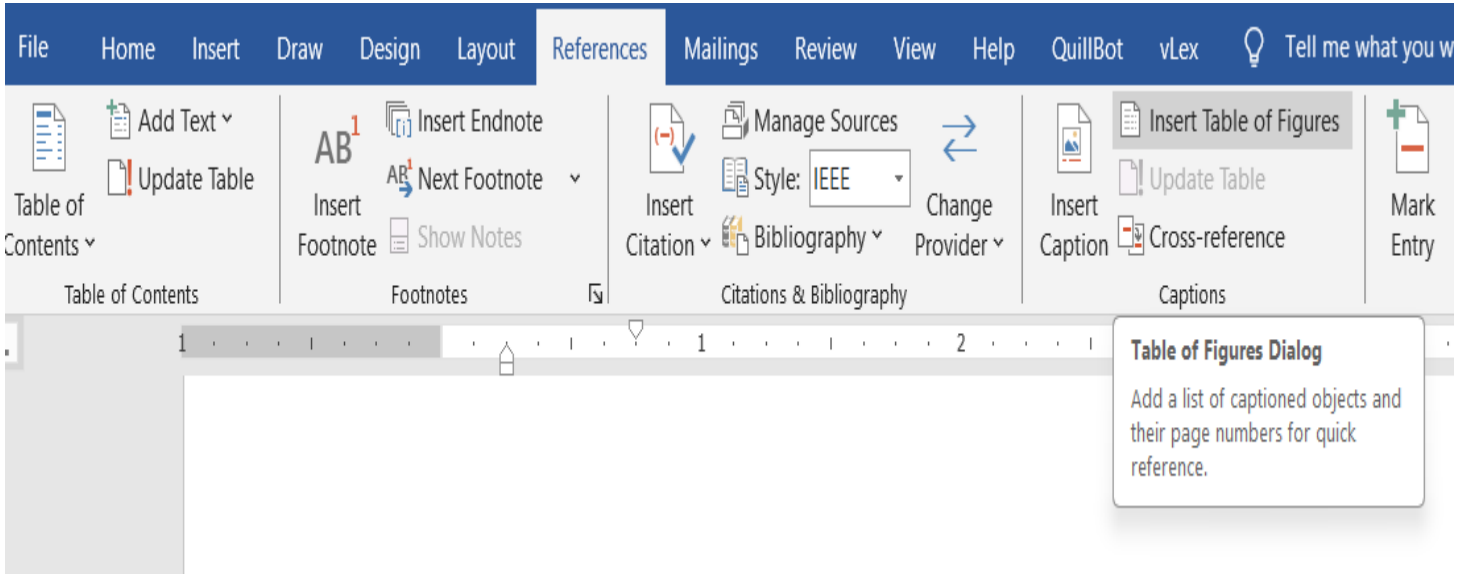


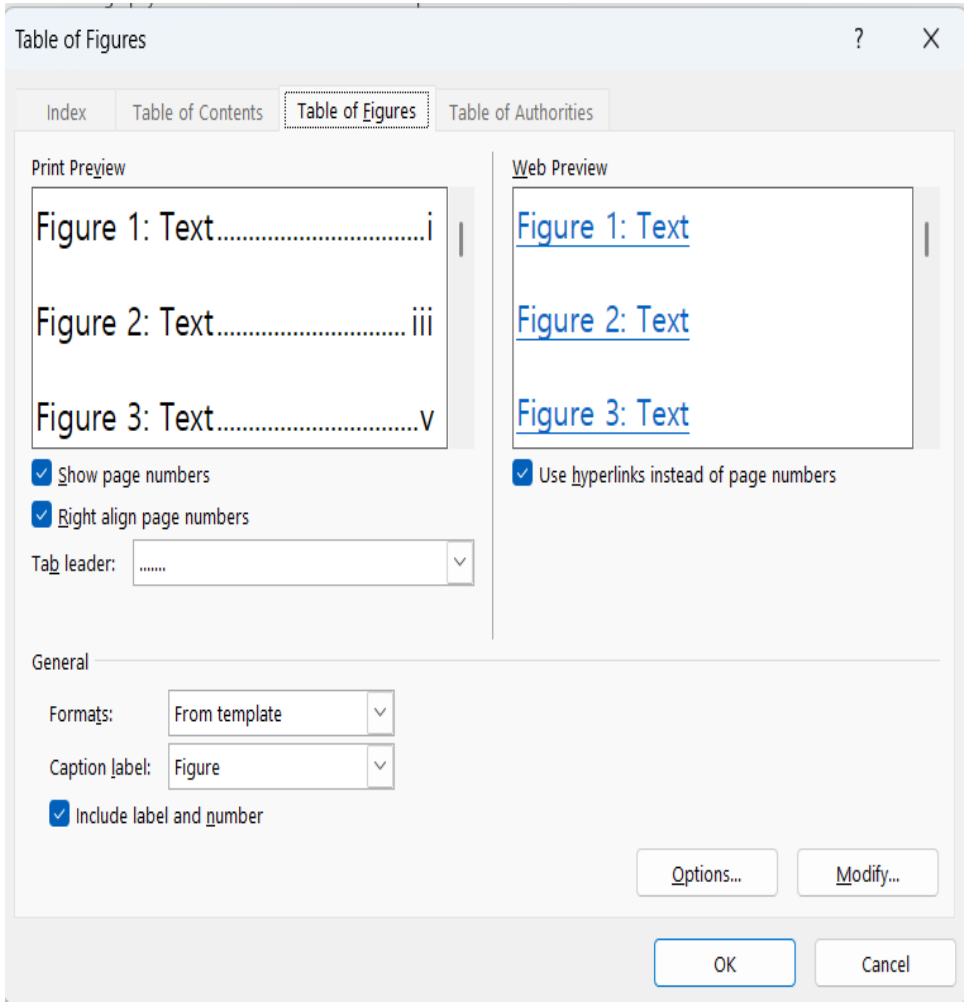




➢ How to insert a Table of Tables and Table of Figures in Word?

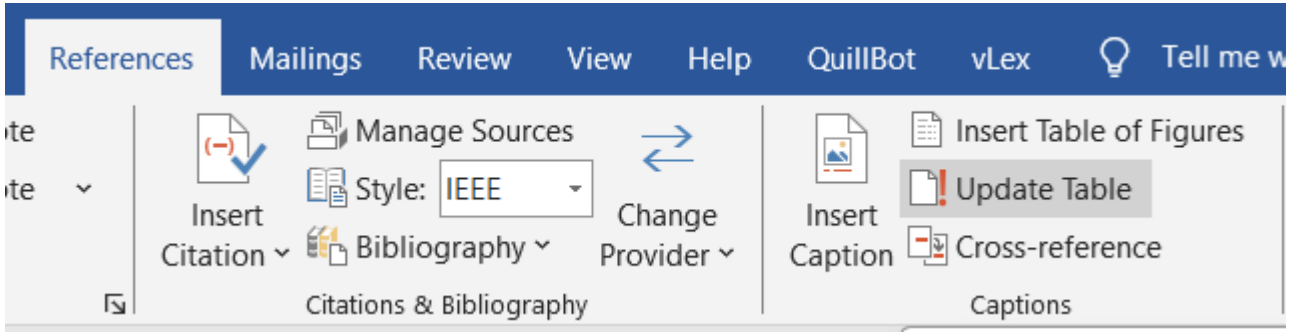
- To create a table of tables or table of figures in Word, you must first add captions to all your tables and figures using the Insert Caption feature on the References tab.
- Then, place your cursor where you want the list to appear, go to the **References tab**, click **Insert Table of Figures**, select the appropriate caption label (Table or Figure), choose your desired format, and **click OK**.

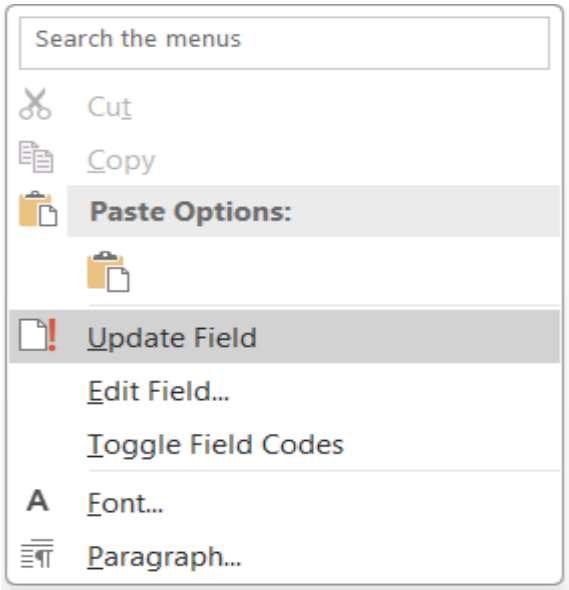


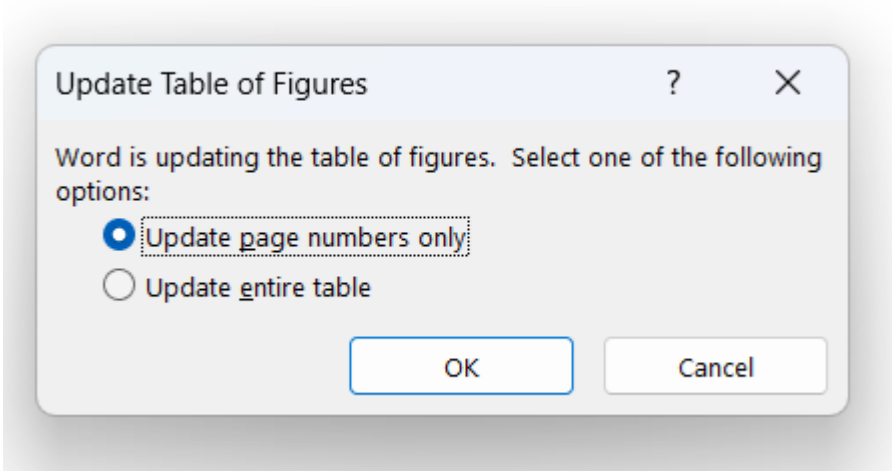


➢ How to update Table of Tables and Table of Figures in Word?

- To update a table of tables or figures, select the table of tables or figures, then go to the **References tab** and click **Update Table** to choose to update the page numbers or the entire table.
- Alternatively, you can **right-click the table** and select **Update Field** to get the same dialog box.







10- Equations

- All equations must be centered and numbered,
- All equations must be referred to in the text (e.g. ‘as explained by eqn. (1)’); avoid using words such as ‘equation above’.

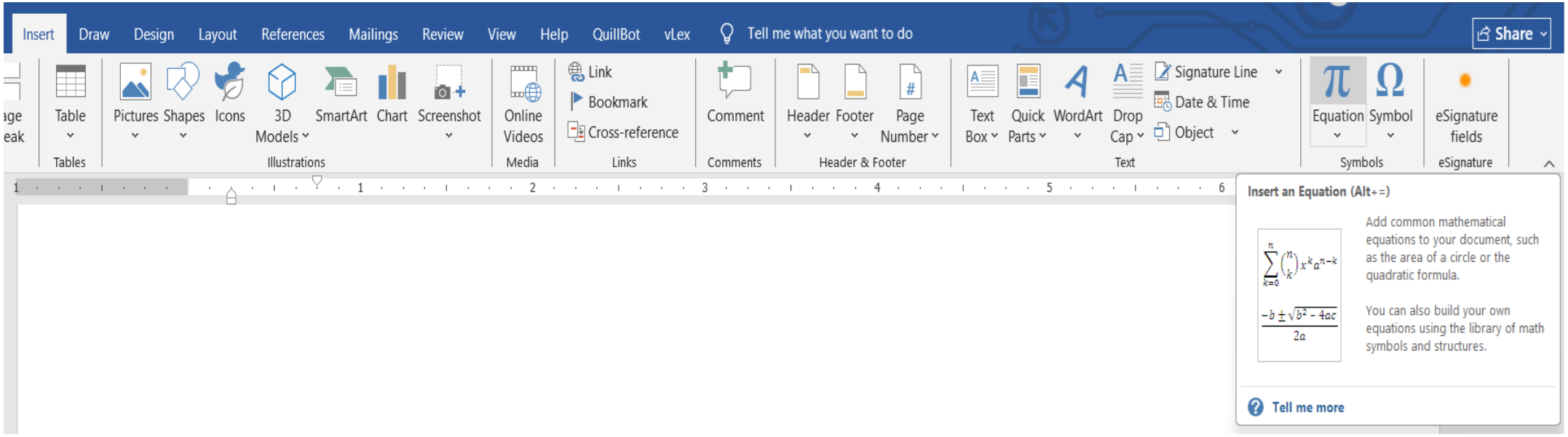
Example:

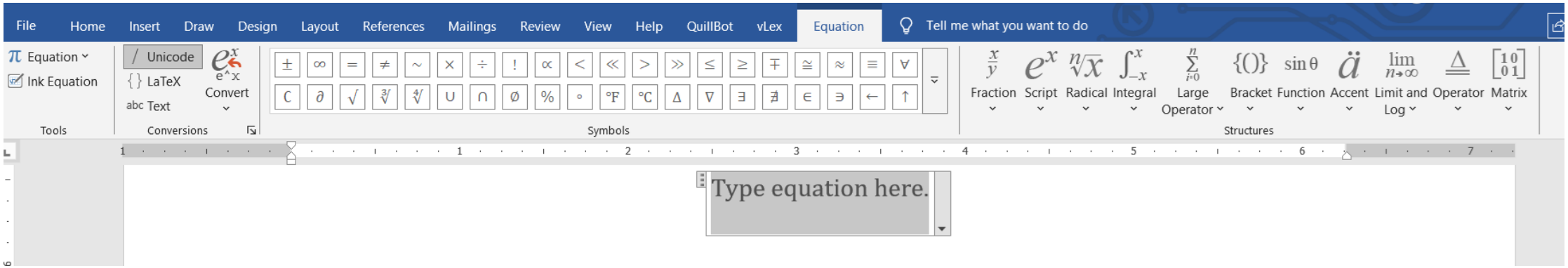
The current (I) and voltage (V) in the single-diode model are related as defined by eqn. (1).

$$I = I_{ph} - I_o \left[exp\left(\frac{V + IR_s}{V_T}\right) - 1 \right] - \frac{V + IR_s}{R_{sh}} \qquad (1)$$

➢ How to insert Equations in Word?

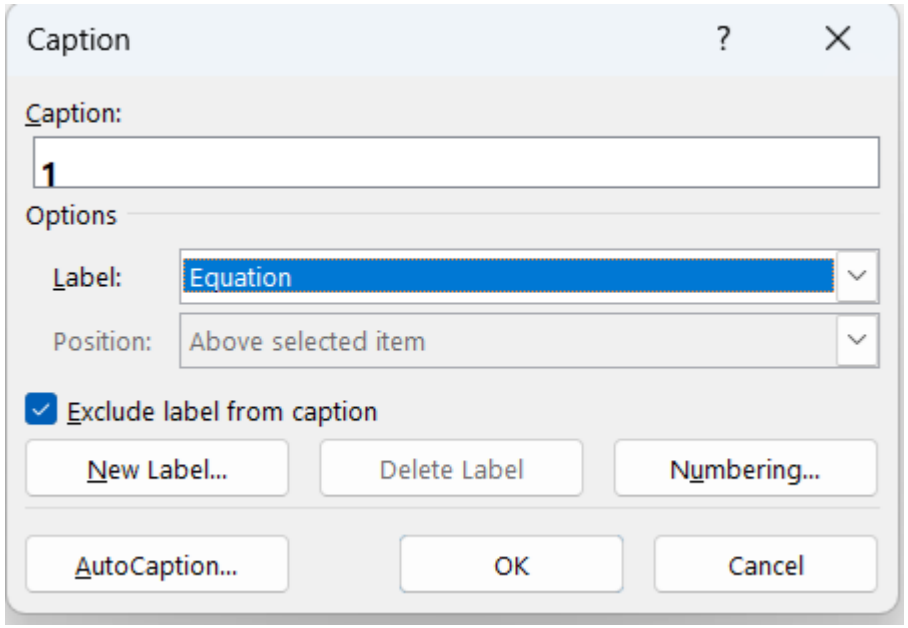
- To insert an equation in Word, go to the **Insert tab**, then select **Equation** from the Symbols group.
- You can then either choose a pre-made equation from the gallery or select Insert New Equation to build your own using the equation editor, which includes tools for symbols and structures like fractions and scripts.

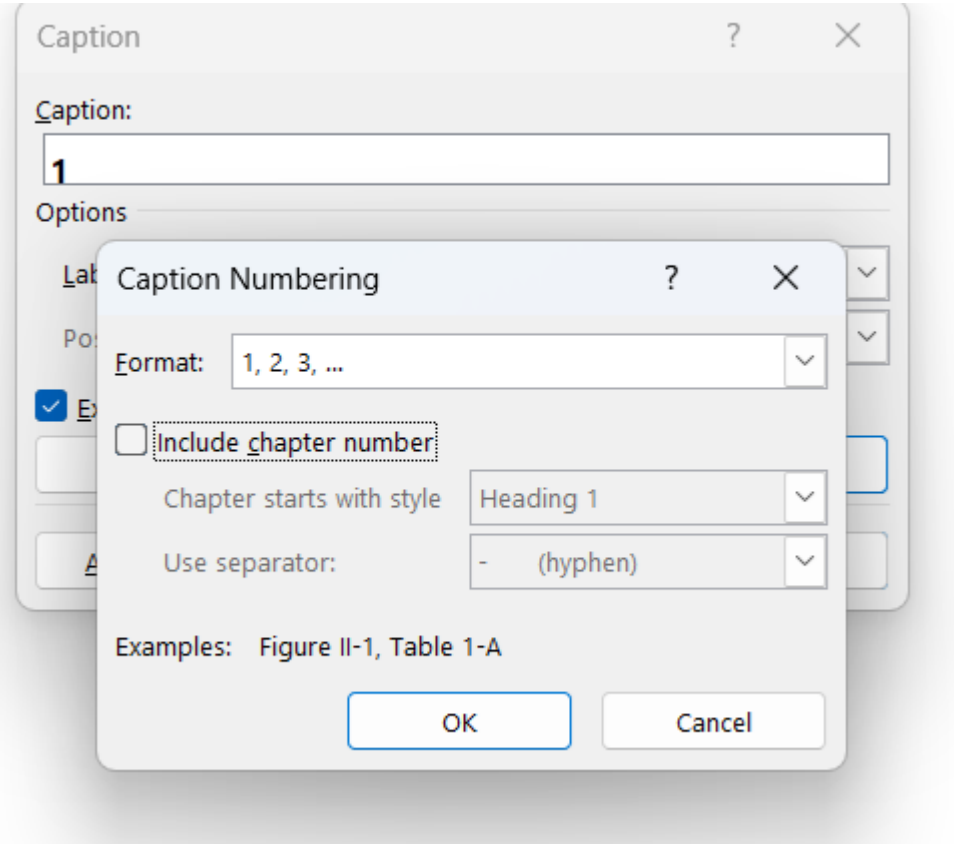




➢ How to numbering equation in Word?

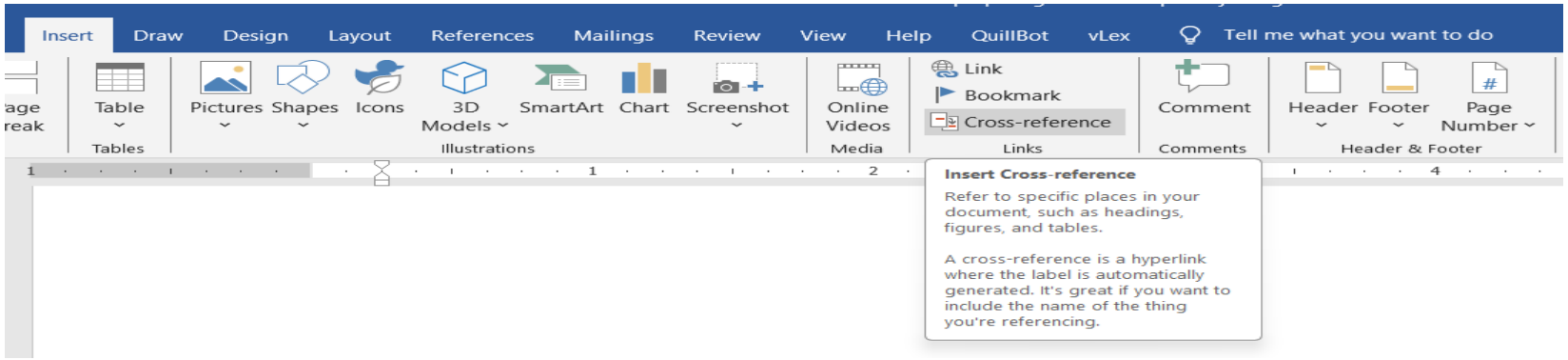
- First insert the equation, then go to the **References tab**, click **Insert Caption**, select "Equation" as the Label, choose the position, and click OK. This automatically adds a sequential number like "(1)" next to your equation.
- You can customize the numbering format, including adding chapter numbers, by clicking the Numbering button in the Caption dialog box.

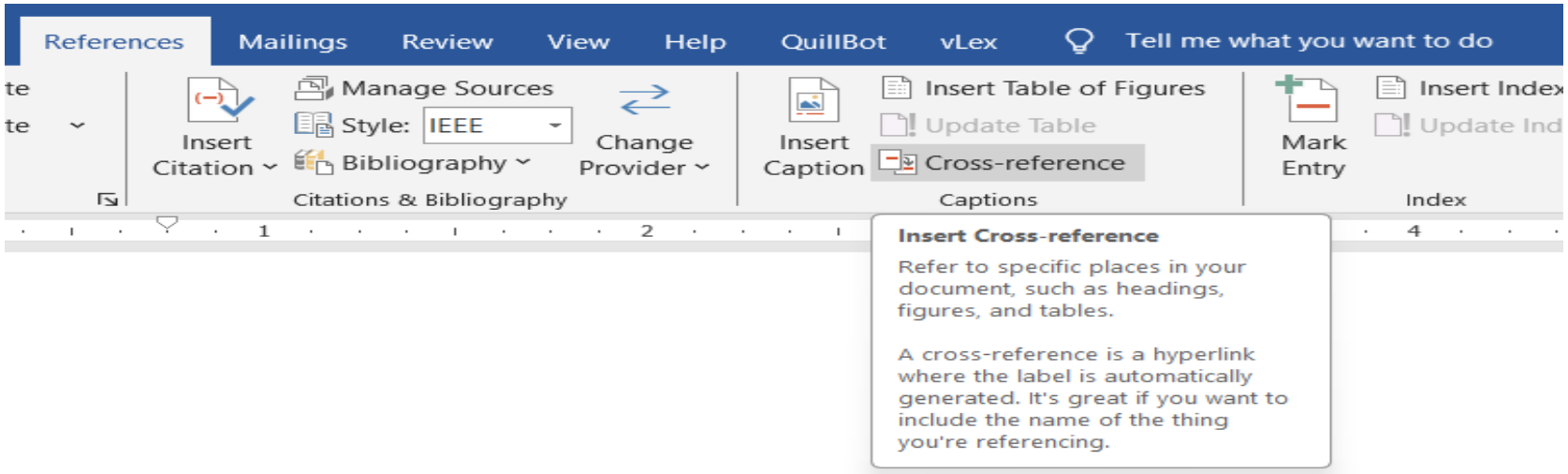


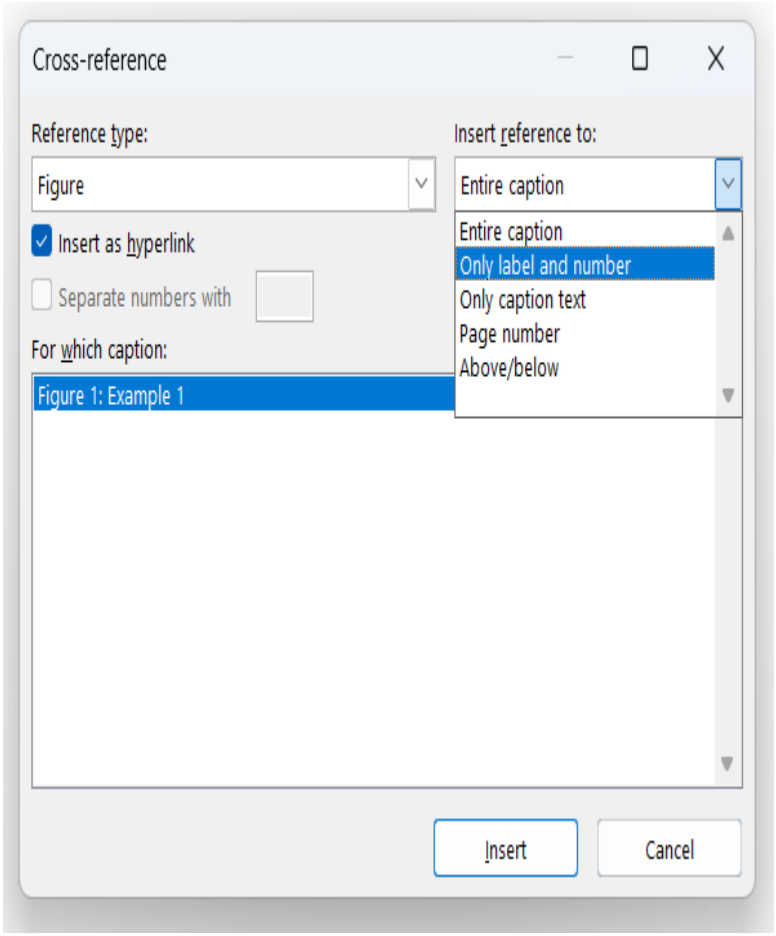


➢ How to refer equation or table or figure in text?

Place the mouse cursor to where you want to insert the figure/table/equation number. Click on "**Insert**" -> "**Reference**" -> "**Cross-reference**". In the dialog box, select "Figure or Table or Equation" as "Reference type", then select "Only label and number", pick "Figure 1 … or Table 1 … or Equation 1 … ", click "**Insert**".



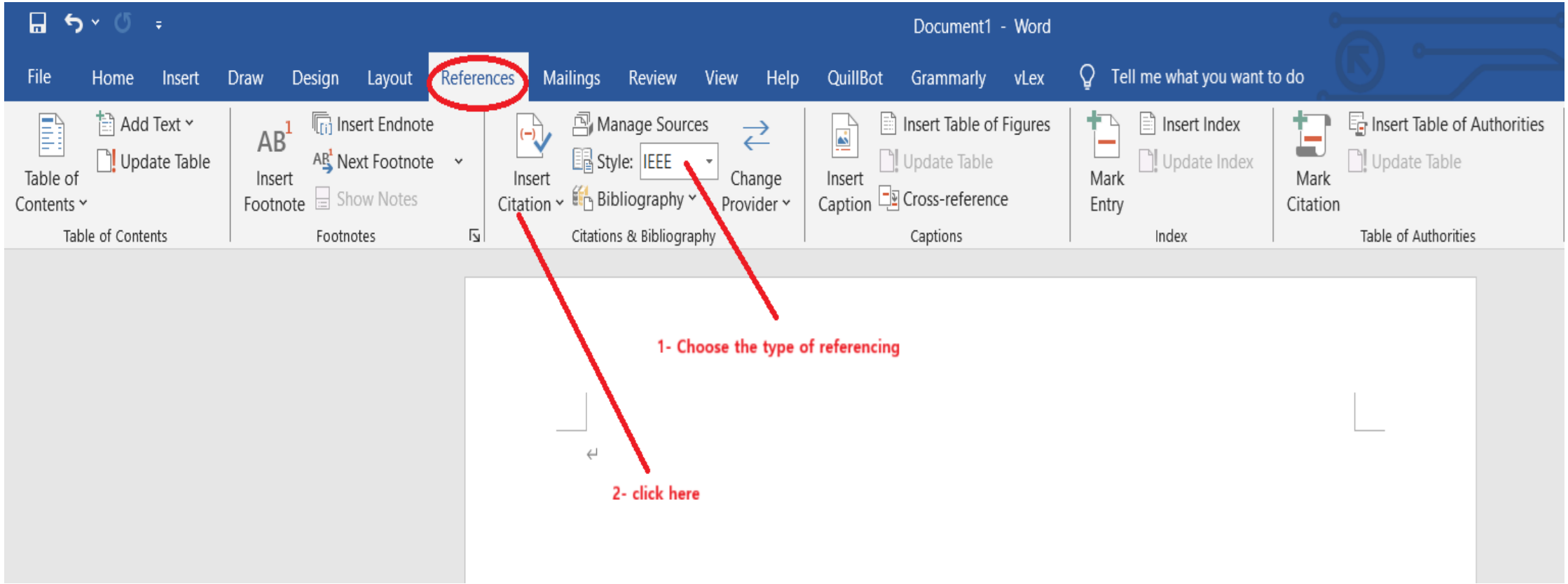




11- Citations and References

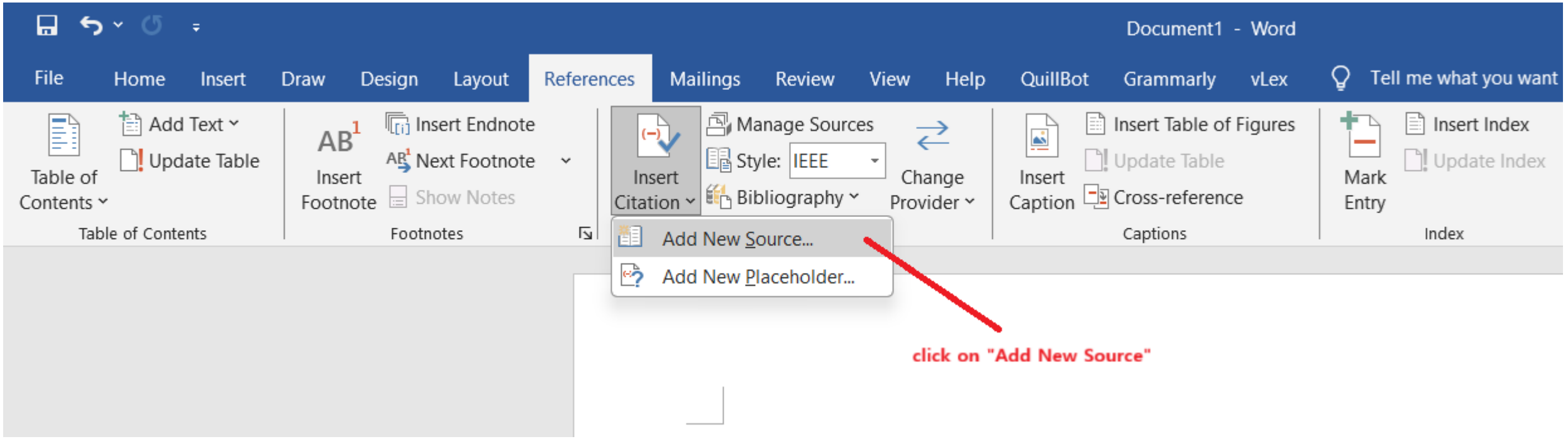
➢ How to insert citations in Word?

- To insert a citation in Word, first, click on the **References tab**, then select your desired citation style from the Style drop-down menu.

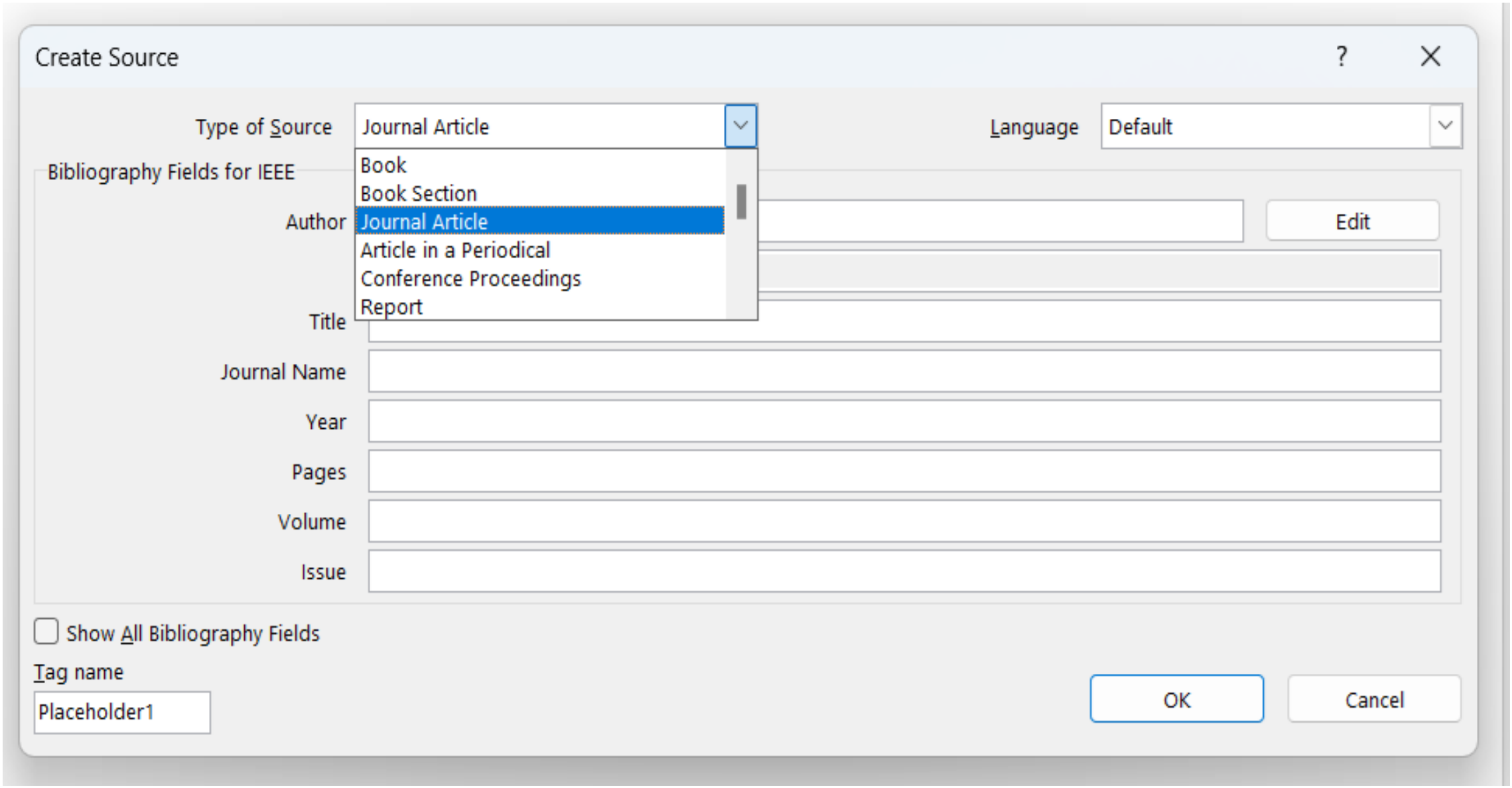


➢ How to insert citations in Word? … Cont.

- Place your cursor at the end of the sentence you want to cite, and click the **Insert Citation** button. From the drop-down, you can either choose an existing source or select Add New Source to create a new one by filling in the required details in the pop-up box.

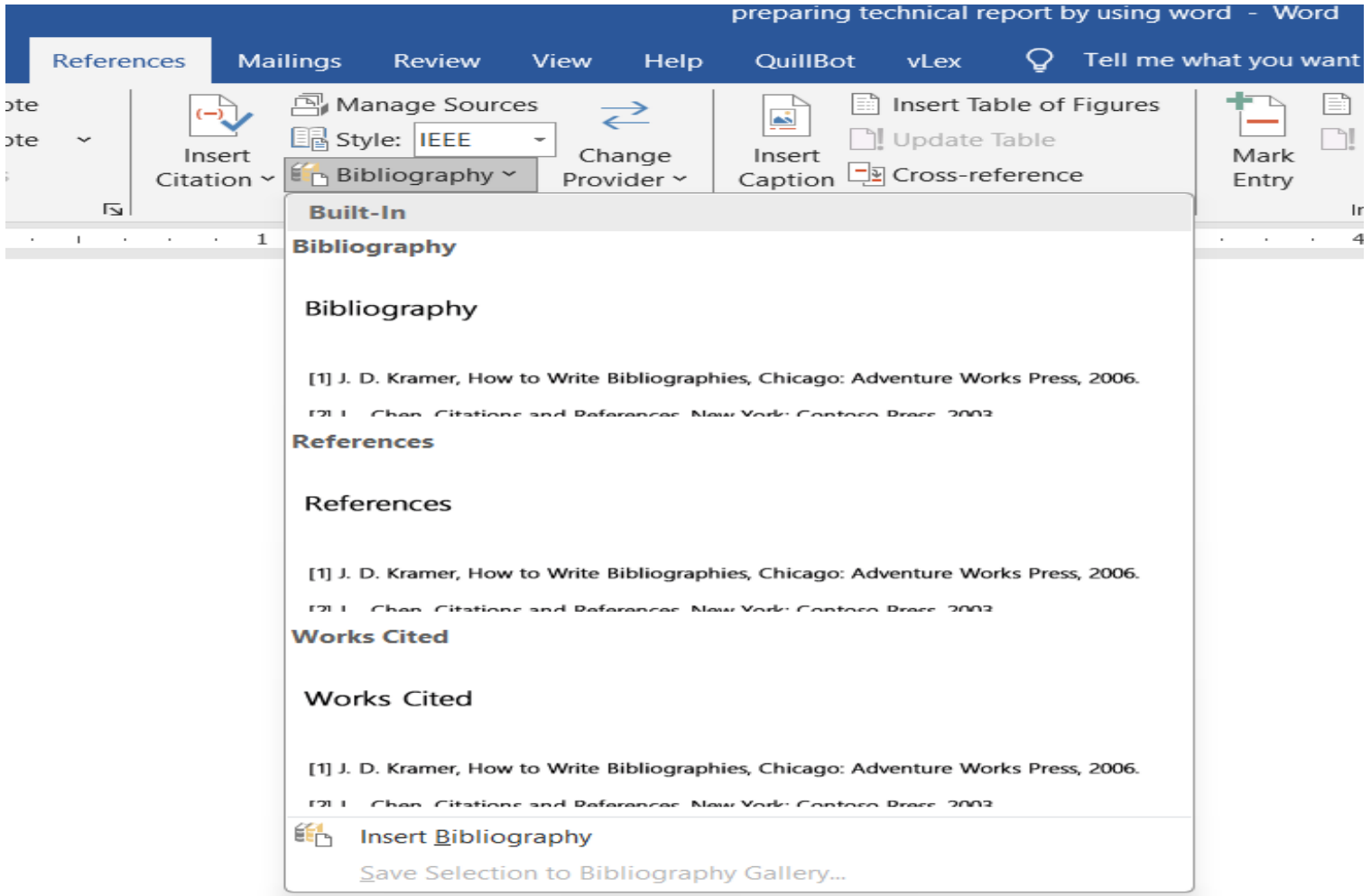


➢ How to insert citations in Word? … Cont.



➢ How to add references in word?

- Once you have added all your sources, place your cursor at the end of your document and click **References** > **Bibliography** > References to automatically generate your bibliography.



End of Lecture